

CONSTRUCTOR UNIVERSITY

School of Computer Science and Engineering

CaML-OP: Causal Machine Learning for Oncology Personalization

*A Hybrid Predictive–Causal Framework for Explainable Oncology,
Informed by the EU AI Act*

by

Muhammad Owais Suhail

Bachelor Thesis in Computer Science

Supervisor: Ishansh Gupta

Co-Supervisor: Prof. Dr. Hendro Wicaksono

Submission: 19 May 2026

CONSTRUCTOR UNIVERSITY

School of Computer Science and Engineering

CaML-OP: Causal Machine Learning for Oncology Personalization

A Hybrid Predictive-Causal Framework for Explainable Oncology, Informed by the EU AI Act

by

Muhammad Owais Suhail

Bachelor Thesis in Computer Science

Supervisor: Ishansh Gupta

Co-Supervisor: Prof. Dr. Hendro Wicaksono

Submission: 19 May 2026

Statutory Declaration

Family Name, Given/First Name	Suhail, Muhammad Owais
Matriculation number	30006217
Kind of thesis submitted	Bachelor Thesis

English: Declaration of Authorship

I hereby declare that the thesis submitted was created and written solely by myself without any external support. Any sources, direct or indirect, are marked as such. I am aware of the fact that the contents of the thesis in digital form may be revised with regard to usage of unauthorized aid as well as whether the whole or parts of it may be identified as plagiarism. I do agree my work to be entered into a database for it to be compared with existing sources, where it will remain in order to enable further comparisons with future theses. This does not grant any rights of reproduction and usage, however.

This document was neither presented to any other examination board nor has it been published.

German: Erklärung der Autorenschaft (Urheberschaft)

Ich erkläre hiermit, dass die vorliegende Arbeit ohne fremde Hilfe ausschließlich von mir erstellt und geschrieben worden ist. Jedwede verwendeten Quellen, direkter oder indirekter Art, sind als solche kenntlich gemacht worden. Mir ist die Tatsache bewusst, dass der Inhalt der Thesis in digitaler Form geprüft werden kann im Hinblick darauf, ob es sich ganz oder in Teilen um ein Plagiat handelt. Ich bin damit einverstanden, dass meine Arbeit in einer Datenbank eingegeben werden kann, um mit bereits bestehenden Quellen verglichen zu werden und dort auch verbleibt, um mit zukünftigen Arbeiten verglichen

werden zu k "onnen. Dies berechtigt jedoch nicht zur Verwendung oder Vervielf "altigung.

Diese Arbeit wurde noch keiner anderen Pr "ufungsbeh "orde vorgelegt noch wurde sie

bisher ver "offentlicht.

19 May 2026 Owais

Date, Signature

Acknowledgements

First and foremost, I want to express my sincere gratitude to my supervisor, Ishansh Gupta, for his invaluable guidance and technical feedback throughout the development of CaML-OP. His patience through the countless iterations of this pipeline truly shaped the final form of this work.

I am equally grateful to my co-supervisor, Prof. Dr. Hendro Wicaksono, for his oversight of the academic process and his sharp, constructive criticism of the regulatory framing.

My thanks also go to the faculty and staff at the School of Computer Science and Engineering at Constructor University for providing the academic environment that made this thesis possible.

Finally, I gratefully acknowledge The Cancer Genome Atlas (TCGA) Research Network and the National Cancer Institute for making the TCGA-OV dataset publicly available. This work would also not be possible without the incredible open-source communities behind EconML, XGBoost, scikit-learn, lifelines, and the broader Python scientific ecosystem.

Abstract

Tailoring cancer treatments to individual patients remains a major challenge in oncology. Right now, decisions about platinum-based chemotherapy for ovarian cancer are mostly based on broad population averages. This approach completely misses the huge differences in how individual patients respond to treatment. To make matters worse, most clinical AI tools predict outcomes rather than identifying actual causes, and almost all of them were built before the EU Artificial Intelligence Act was introduced.

This thesis introduces CaML-OP (Causal Machine Learning for Oncology Personalization). It is a hybrid predictive causal framework that pairs a gradient-boosted prognostic encoder (XGBoost) with R- and Doubly Robust (DR) causal meta-learners. We also integrated an EffectSHAP attribution layer, an LLM narrative module featuring automated faithfulness checks, and a functional Streamlit dashboard prototype. We designed this dashboard with the EU AI Act’s high-risk AI requirements in mind using patient-level explanations and subgroup reporting as practical design features to address those rules rather than treating them as strict legal mandates.

We evaluated the pipeline using TCGA-OV covariates (leaving a working sample of $n = 547$ after preprocessing) across a semi-synthetic benchmark with four data-generating processes and ten runs per scenario. We tested it against six causal baselines and three binary-outcome predictive comparators. To confirm the architecture was stable and statistically consistent, we ran a purely synthetic verification at N up to 4000.

Using the off-policy AIPW estimator, CaML-OP hit the highest mean estimated policy value ($V = 0.559$). However, because the estimator ranks the "treat-all" and "oracle" benchmarks very close together, this result means that CaML-OP's estimated value is not statistically distinguishable from the oracle benchmark under this finite-sample off-policy estimator (Wilcoxon $p = 0.42$), rather than proving it is fully equivalent to knowing the true treatment effects. Notably, CaML-OP is the only ensemble to provide an approximate patient-level uncertainty interval; its empirical coverage on the test set reached 0.847 against a nominal 0.95 target. Subgroup PEHE showed only minor variations across age, FIGO stage, and race strata, indicating no major error concentration on this benchmark, though this does not constitute a full fairness audit.

Ultimately, the core contribution of this work is at the system level. We have built a unified framework that weaves together XGBoost leaf embeddings, R/DR ensembles, causal explainability, and EU AI Act-informed design to estimate individual treatment effects in ovarian cancer

from routine clinical data. CaML-OP is presented strictly as a research prototype, not a deployable clinical tool.

Keywords: *Causal Machine Learning; Individualized Treatment Effects; Explainable AI; Ovarian Cancer; Platinum Chemotherapy; EU AI Act; XGBoost; Clinical Decision Support.*

List of Abbreviations

Abbreviation	Meaning
AIPW	Augmented Inverse Probability Weighting
ATE	Average Treatment Effect
AUC	Area Under the ROC Curve
AUPRC	Area Under the Precision-Recall Curve
CaML-OP	Causal Machine Learning for Oncology Personalization
CDSS	Clinical Decision Support System
CI	Confidence Interval
Cox PH	Cox Proportional Hazards model
DAG	Directed Acyclic Graph
DGP	Data-Generating Process
DML	Double Machine Learning
DR	Doubly Robust
EU AI Act	Regulation (EU) 2024/1689 - the EU Artificial Intelligence Act
FIGO	International Federation of Gynecology and Obstetrics
GDC	Genomic Data Commons
HGSOC	High-Grade Serous Ovarian Cancer
HTE	Heterogeneous Treatment Effect
IARC	International Agency for Research on Cancer
ITE	Individualized Treatment Effect
LLM	Large Language Model
MAE	Mean Absolute Error
MICE	Multivariate Imputation by Chained Equations
NCI	National Cancer Institute
PDX	Patient-Derived Xenograft
PEHE	Precision in Estimation of Heterogeneous Effects
RCT	Randomized Controlled Trial
RF	Random Forest
RSF	Random Survival Forest
SHAP	SHapley Additive exPlanations
SUTVA	Stable Unit Treatment Value Assumption
TCGA-OV	The Cancer Genome Atlas - Ovarian Serous Cystadenocarcinoma

Abbreviation	Meaning
XAI	Explainable Artificial Intelligence
XGBoost	Extreme Gradient Boosting

Contents

Statutory Declaration	ii
Acknowledgements	iv
Abstract	v
List of Abbreviations	vii
1 Introduction	1
1.1 The clinical problem	1
1.2 Why existing AI systems fall short	2
1.3 What this thesis contributes	2
1.4 Research questions	3
1.5 Thesis structure	4
2 Related Work	5
2.1 Predictive survival modelling	5
2.2 Causal ML for treatment effect heterogeneity	5
2.3 Benchmarking ITE estimators	6
2.4 Explainability and the EU AI Act	6
2.5 LLMs in clinical settings	7
2.6 Platinum response heterogeneity in HGSOE	7
2.7 Positioning	7
3 Methodology	9
3.1 Pipeline overview	9
3.2 The dual role of XGBoost	9
3.3 Data preprocessing	10
3.3.1 Missing data	10
3.3.2 Splitting strategy	10
3.4 XGBoost leaf encoder	10
3.5 Causal estimation	11
3.5.1 Identification assumptions	11
3.5.2 R-Learner	11

3.5.3	DR-Learner	12
3.5.4	Ensemble and uncertainty reporting	12
3.6	Explainability layer	12
3.6.1	Effect-SHAP	12
3.6.2	LLM narrative and faithfulness checks	12
3.7	Clinical dashboard prototype	13
3.8	Semi-synthetic evaluation framework	13
3.8.1	Why semi-synthetic	13
3.8.2	Four DGP scenarios	14
4	Experimental Setup	15
4.1	Dataset and cohort derivation	15
4.2	Implementation	17
4.3	Baselines	17
4.4	Evaluation metrics	18
4.5	Synthetic verification	19
4.6	Reproducibility	19
5	Results	20
5.1	Overview	20
5.2	ITE accuracy and policy quality	20
5.3	Binary-outcome predictive comparators	23
5.4	Uncertainty interval coverage	24
5.5	Subgroup analysis	25
5.6	Synthetic verification	26
5.6.1	Check 1 - Correctness	26
5.6.2	Check 2 - Competitiveness under strong confounding	27
5.6.3	Check 3 – Consistency	28
5.7	Summary	29
5.8	Statistical significance	29
6	Discussion	31
6.1	What the results show	31
6.2	Why the ensemble works	31
6.3	Answering the research questions	32
6.3.1	RQ1 - Heterogeneity of platinum effects	32
6.3.2	RQ2 - LLM narrative fidelity	32
6.3.3	RQ3 - Fairness across subgroups	32
6.4	Relation to prior work	33
6.5	Limitations	33
6.6	Ethical considerations	34
6.6.1	Automation bias and clinician oversight	34
6.6.2	Biased recommendations and underrepresented groups	34

6.6.3	Binary treatment and clinical completeness	34
6.6.4	Patient privacy	34
6.6.5	Non-deployment status	34
6.7	Threats to validity	35
6.7.1	Internal validity	35
6.7.2	External validity	35
6.7.3	Construct validity	35
6.7.4	Conclusion validity	35
7	Conclusion	36
	References	38

List of Figures

3.1	CaML-OP full pipeline. Inputs (TCGA-OV covariates, $n = 547$ after preprocessing) enter on the left. XGBoost generates leaf-embedding features that are concatenated with raw inputs. These pass to an R-Learner (NonParamDML with 5-fold cross-fitting) and a DR-Learner (ForestDRLearner) in parallel, whose outputs are averaged to produce ITE τ -hat. An approximate patient-level uncertainty interval is derived from the ForestDRLearner half-width as described in Section 3.5.4. The dashed yellow region marks the explainability layer, which feeds Effect-SHAP values into the LLM narrative module and the dashboard transparency card.	9
3.2	Causal DAG for the platinum therapy analysis. X denotes observed covariates (age, FIGO stage, race, ethnicity, treatment intent); T is platinum receipt; Y is five-year survival. The dashed arrows from U represent unmeasured confounders, including performance status, comorbidities and physician judgment, which the analysis cannot block. $\tau(X)$ is the individualized treatment effect of interest.	11
3.3	CaML-OP dashboard prototype mockup. Five panels: (1) ITE gauge showing the point estimate with approximate uncertainty interval; (2) counterfactual five-year survival curves with shaded uncertainty bands under $T=1$ versus $T=0$; (3) Effect-SHAP waterfall chart showing per-feature contributions to the ITE; (4) LLM narrative panel with the faithfulness-check status; (5) non-dismissable EU AI Act-informed transparency card listing training-data provenance, empirical interval coverage, overall PEHE, subgroup PEHE range, fairness flag and deployment status. The card explicitly states RESEARCH PROTOTYPE ONLY.	13
3.4	Scatter plots of $\tau(X)$ against standardized age for the four DGP scenarios. S_1 is flat at $\tau = 0.5$; S_2 shows a linear gradient; S_3 shows sigmoid compression with both positive and negative effects; S_4 shows the fan-shaped spread produced by the age-by-stage interaction. Colour from red through yellow to green encodes τ magnitude from negative to positive.	14
4.1	TCGA-OV cohort derivation flowchart from the original GDC pull through outcome, treatment and covariate filters to the working sample $n = 547$	16

5.1	Grouped bar chart of mean PEHE by DGP scenario for each of the seven evaluated causal methods. Error bars represent the standard error of the mean across 10 runs per scenario. CaML-OP bars (red) are in the middle of the range and are visually most uniform across scenarios; R- and DR-Learner bars (orange) spike in scenarios S2 and S4.	21
5.2	Pooled PEHE, MAE and ATE error side by side. Each panel sorts methods from best (left) to worst (right). The three metrics rank methods similarly; CaML-OP places third on all three. Error bars are standard errors over 40 runs.	22
5.3	Off-policy estimated AIPW policy value with 95% error bars ($1.96 \times \text{SE}$). CaML-OP is the leftmost bar in red. The green bar marks the oracle benchmark. The dashed horizontal line marks the oracle value at 0.548. Reference policies TreatAll and TreatNone shown in grey at the right.	23
5.4	Three-panel bar chart showing Cox PH (fixed-horizon), RF classifier and XGBoost classifier on AUC, AUPRC and Brier score. Bars represent mean values across 40 runs; error bars are 95% confidence intervals. Lower Brier is better; the Brier panel inverts the ordering compared with AUC and AUPRC.	24
5.5	Two-panel chart. Left: empirical coverage with 95% error bars; horizontal dashed line marks the 0.95 nominal target. Right: mean interval width across the test set. Both methods under-cover; CaML-OP's interval is slightly wider but its coverage is somewhat lower than the Causal Forest's.	25
5.6	Subgroup PEHE heatmap with methods on rows and seven subgroups on columns. Darker shading indicates higher PEHE. CaML-OP's row (third from top) is visibly more uniform than R-Learner and DR-Learner rows. Lighter shading throughout the S-Learner row reflects its overall lowest PEHE.	26
5.7	Two-panel comparison of synthetic Checks 1 and 2. Left panel (Check 1): standard confounding; horizontal dashed line marks the $\text{std}(\tau_{\text{true}}) = 0.097$ pass threshold. Right panel (Check 2): doubled confounding. Bars sorted by PEHE within each panel. Error bars are standard errors over 8 runs.	27
5.8	PEHE versus training N on a log-scaled x-axis. CaML-OP's line (red, bold) shows the steepest descent. Shaded bands represent ± 1 standard deviation across 5 runs per N. Lower is better. All methods except the S-Learner show monotone decreasing PEHE.	28
5.9	Effect-SHAP attribution. Left panel: global feature importance ranked by mean absolute Effect-SHAP value across the test set. Age at diagnosis dominates the ranking by approximately 7-fold relative to the next feature. Right panel: single-patient waterfall for a 58-year-old patient with FIGO stage IV and $\tau\text{-hat} = +0.236$; bar lengths represent each feature's contribution to the patient's deviation from the mean ITE. Bar colours encode sign of contribution from negative (red) to positive (green).	30

List of Tables

2.1	Comparison of related methods. Causal = uses a causal identification strategy; Hybrid = combines a predictive encoder with a causal estimator; XAI = per-patient causal explanation; EU AI Act = designed with reference to Regulation (EU) 2024/1689 obligations; TCGA-OV = evaluated on TCGA-OV.	8
4.1	TCGA-OV variables used in CaML-OP after cleaning.	16
4.2	Baseline methods grouped by type.	17
4.3	Evaluation metrics.	18
5.1	Mean PEHE, MAE and ATE error pooled over 4 DGP scenarios x 10 runs (n_{test} approximately 137 per run). Methods sorted by PEHE.	20
5.2	Per-scenario PEHE (mean over 10 runs per scenario). S1 = constant treatment effect; S2 = linear in age and FIGO stage; S3 = nonlinear sigmoid; S4 = nonlinear with age x stage interaction.	21
5.3	Off-policy estimated AIPW value of policy $\pi(X) = 1\{\hat{\tau}(X) > 0\}$, pooled over 4 scenarios x 10 runs. Higher is better. Oracle benchmark = 0.548. Standard deviation across 40 runs reported.	22
5.4	Binary-outcome predictive comparators: AUC, AUPRC, Brier score (mean over 4 scenarios x 10 runs). Lower Brier is better; higher AUC and AUPRC are better. 23	
5.5	Empirical coverage of the reported 95%-nominal-target intervals on the TCGA-OV semi-synthetic test set (n_{test} approximately 137 per run, pooled over 4 scenarios x 10 runs). Coverage = fraction of true τ values inside the reported interval. SD = standard deviation across 40 runs. Mean width = average interval width across the test set. Nominal target = 0.95.	24
5.6	Subgroup PEHE by age band, FIGO stage, and race grouping (mean over 4 scenarios x 10 runs). Subgroup definitions: age <50 , $50-65$, ≥ 65 ; stage Low/High split at median; race Majority/Minority based on most common category versus all others.	25
5.7	Synthetic Check 1 - architectural correctness. $N = 2000$, 8 runs, Scenario 4, standard confounding ($\eta = 1$).	26
5.8	Synthetic Check 2 - competitiveness under strong confounding. $N = 2000$, 8 runs, Scenario 4, doubled confounding ($\eta = 2$).	27

5.9 Synthetic Check 3 – statistical consistency. Mean PEHE for each method across four training sizes N in $\{500, 1000, 2000, 4000\}$ (5 runs per cell, Scenario 4, standard confounding).	28
---	----

Chapter 1

Introduction

1.1 The clinical problem

Ovarian cancer continues to pose a major challenge to global health. Data from the IARC GLOBOCAN report establishes a heavy global baseline with an estimated 324,603 women diagnosed with the disease worldwide in a single year (Bray et al., 2024) and long-term projections estimating a near 47% increase in global burden by 2050. Because symptoms are often quiet at first, most patients face an advanced diagnosis by the time they see a doctor. This late detection drives down survival rates and places an intense burden on initial treatment decisions. Historically the standard approach relied on primary cytoreductive surgery followed by adjuvant chemotherapy. However recent medical consensus highlights a major practice shift toward using neoadjuvant chemotherapy giving a combination of carboplatin and paclitaxel *before* interval surgery for advanced cases to improve management and surgical outcomes (ASCO, 2025). Yet, how patients respond to these regimens remains incredibly varied. While some individuals experience complete remission, others see their disease progress within months meaning they endure severe side effects for little to no clinical return.

The core problem this thesis tackles is not a biological mystery but an informational one. Right now clinical guidelines depend on population averages from clinical trials. By design these averages cannot predict how a specific individual will react to a drug. While a solid prognostic model can tell us *who* is likely to have a good outcome, it cannot prove whether the treatment itself actually caused that success (Obermeyer and Emanuel, 2016). This gap between prediction and causation has real world consequences for patient well-being. For instance, Prigerson et al. (2015) demonstrated that aggressive chemotherapy near the end of life frequently erodes a patient's quality of life without offering any survival advantage. Wright et al. (2016) noted similar clinical missteps when doctors lacked specific information on individual treatment effects.

Precision oncology has tried to bridge this divide through functional testing, using tools like patient derived xenografts (Gao et al., 2015) and organoid assays (Pauli et al., 2017). The science behind these methods is promising but practical access is restricted to a handful of elite academic medical centers. A patient living in a rural area or relying on a regional hospital

rarely has a realistic path to organoid guided care, a disparity that worsens along racial and geographic lines. Furthermore, these inequities are baked into the data itself Landry et al. (2018) highlighted that patients of African descent are systematically left out of the genomic databases used to drive modern precision medicine.

This thesis builds on the idea that personalized cancer care should not require access to cutting-edge research facilities. This need is especially urgent now that treatment decisions are being made earlier in the care cycle, before surgical tissue is even available. If the basic clinical data already gathered by everyday oncology practices contains enough of a signal to capture how individual treatment responses differ then developing a causal inference pipeline around those existing variables is a deeply worthwhile pursuit.

1.2 Why existing AI systems fall short

Machine learning has made clear strides in predicting ovarian cancer outcomes over the last few years. Tools like XGBoost classifiers, Random Survival Forests, and standard Cox regression models have proven they can group patients into distinct risk categories quite accurately using validation datasets. In fact a few of these models are finally stepping out of the lab and into clinical trials to help screen patients for eligibility.

The real issue is not the math itself, but how we apply these systems. When a model predicts which patients are likely to live longer, it is answering a completely different question than whether a specific therapy will actually help a specific individual. The first scenario is observational, the second is causal. If we mistake a survival prediction for a guarantee of treatment benefit, we risk introducing severe algorithmic bias. This is exactly the kind of systemic flaw Obermeyer et al. (2019) uncovered when they looked at a massive, widely deployed healthcare risk model that inadvertently penalized vulnerable patient groups.

Another major hurdle is trust and explainability. Doctors are notoriously slow to adopt AI recommendations and for good reasons. They cannot look under the hood to see how a model reached its conclusion. On top of that, the legal landscape has completely shifted. Under Regulation (EU) 2024/1689—better known as the EU AI Act any AI system used for clinical decision support is now officially labeled a "high-risk" application (European Parliament and Council, 2024). This means developers can no longer just throw a model together. They are legally bound to strict rules regarding data governance, technical documentation, transparency and human oversight not to mention rigorous standards for cybersecurity and robustness. Because most existing clinical AI tools were built before these regulations were finalized, very few of them were actually designed to handle this level of accountability from day one.

1.3 What this thesis contributes

CaML-OP combines gradient boosted prognostic encoding with causal meta learning to estimate individualized treatment effects (ITE) for platinum chemotherapy from routine clinical data. It first trains an XGBoost survival model. Extracts its leaf structure to capture non-linear

patterns and then feeds this into an R-Learner and a Doubly Robust Learner to calculate the final averaged ITE.

An explainability layer sits on top. Effect-SHAP isolates the actual causal drivers rather than simple survival predictions. An LLM module translates these metrics into checked clinical summaries.

The interface design aligns with the EU AI Act's high risk framework. Features like per-patient explanations and uncertainty reporting act as design responses to the spirit of the regulation. Making the system AI Act-informed rather than legally deployment-ready.

The remainder of the thesis tests this architecture. A semi-synthetic benchmark pairs real TCGA-OV data with simulated outcomes across four scenarios to evaluate CaML-OP against nine baseline models. Over forty tests are conducted followed by a synthetic verification of statistical consistency.

1.4 Research questions

The main question is whether a hybrid predictive causal pipeline can generate useful individualized treatment-effect estimates for platinum chemotherapy in ovarian cancer using only standard clinical variables.

1.4.1 Objectives

- **O1:** Build a causal ITE estimation pipeline on TCGA-OV data that uses XGBoost leaf embeddings as features for R- and DR-Learner estimators. Then validating it via semi-synthetic benchmarks.
- **O2:** Benchmark CaML-OP against three binary outcome predictive models and five causal meta-learners to evaluate ITE accuracy and off policy value.
- **O3:** Deploy an Effect-SHAP attribution layer and an LLM narrative module with automated accuracy checks.
- **O4:** Show end-to-end functionality through a dashboard prototype featuring subgroup error reporting and a built-in transparency card.

1.4.2 Secondary Questions

- **RQ1:** How varied are the estimated platinum therapy effects across subgroups (age, FIGO stage, race), and which covariates drive treatment benefit the most?
- **RQ2:** Can LLM-generated summaries stay entirely faithful to underlying numeric causal data while remaining easy for clinicians to read?
- **RQ3:** Are there noticeable differences in ITE estimation errors across demographic subgroups on this benchmark that require deeper investigation?

1.5 Thesis structure

Chapter 2 reviews the relevant literature and establishes how CaML-OP builds on existing research. Chapter 3 breaks down the technical components of the pipeline itself. Chapter 4 outlines the experimental design and Chapter 5 present the empirical findings. Finally, Chapter 6 discusses the broader implications of these results, addresses the core research questions and explores ethical concerns alongside threats to validity, leading into concluding remarks in Chapter 7.

Chapter 2

Related Work

2.1 Predictive survival modelling

For decades the Cox proportional hazards model (Cox, 1972) has been the bedrock of oncology survival analysis. It remains popular because hazard ratios are easy to interpret, the model is widely understood and it holds up reasonably well even when its core proportional hazards assumption is slightly violated. Random Survival Forests (Ishwaran et al., 2008) step past this limitation by handling non-linear covariate effects and consistently outperforming Cox models when feature interactions are highly complex. While XGBoost (Chen and Guestrin, 2016) is typically used for standard tabular classification rather than survival tracking. It performs competitively when applied to binary survival endpoints. In fact, Yin et al. (2022) proved its utility specifically for ovarian cancer prognosis.

However, all these methods share a fundamental bottleneck. They are strictly correlative. A survival model tells us that patients with a specific set of traits tend to experience specific outcomes. What it cannot tell us is whether a chosen patient's outcome would actually change if she received platinum therapy versus if she did not. Treating a correlation as a cause can lead to genuine patient harm. A risk starkly illustrated by Obermeyer et al. (2019) when a massive, real-world clinical risk algorithm inadvertently used healthcare spending as a proxy for actual medical need. To address this CaML-OP includes Cox PH and a Random Forest classifier strictly as binary-outcome predictive comparators rather than full time-to-event survival baselines. The exact implications of this decision on construct validity are explored later in Section 6.7.3.

2.2 Causal ML for treatment effect heterogeneity

Traditional meta-learners like the T-Learner, S-Learner, and X-Learner (Künzel et al., 2019) turn this theory into practical code, though their performance varies depending on treatment imbalances and how well the nuisance models are specified.

To handle these challenges more robustly, the R-Learner (Nie and Wager, 2021) leverages Robinson's (1988) partial-linear decomposition to create a Neyman-orthogonal loss function. This

orthogonality is a major advantage. It ensures that first-order mistakes in your nuisance models only cause minor, second-order errors in the final ITE estimate. Similarly, the DR-Learner (Kennedy, 2023) builds in robustness by using doubly robust pseudo-outcomes, which stay accurate as long as either the outcome model or the propensity score model is correct.

CaML-OP combines these two approaches into an equally weighted ensemble. Because each learner handles nuisance errors differently, combining them makes the overall system far more stable across different clinical scenarios. To keep the evaluation grounded, a Causal Forest (Wager and Athey, 2018) is added as a standalone baseline. It uses honest estimation to generate valid confidence intervals making it the perfect benchmark for testing CaML-OP's uncertainty reporting.

2.3 Benchmarking ITE estimators

Curth et al. (2021a) published a vital critique regarding how heterogeneous treatment effect (HTE) benchmarks are typically run. Two of their insights directly shaped the evaluation structure of this thesis. First they noted that published comparisons often evaluate different methods under completely inconsistent conditions which makes it nearly impossible to tell if one algorithm is truly better than another. Second they found that a model's PEHE rank is a poor predictor of how well it actually guides clinical choices. A method with a slightly better PEHE score doesn't automatically mean it will recommend therapies to the right patients. This thesis directly controls for both issues. Finally, while Curth et al. (2021b) also introduced SurvITE specifically to handle time-to-event HTE estimation, this project opts for a binary outcome at a fixed horizon to keep the pipeline practical and mathematically tractable.

2.4 Explainability and the EU AI Act

The EU AI Act officially Regulation (EU) 2024/1689 was adopted in 2024 and officially took effect on August 1st of that year. Under Annex III of the law, AI systems deployed for medical purposes are strictly classified as "high-risk." This label binds developers to rigorous standards regarding data governance, detailed technical documentation, transparency for users, and strong human oversight. This thesis builds entirely on that exact perspective.

On the explainability side, SHAP (Lundberg and Lee, 2017; Lundberg et al., 2020) stands as the gold standard for pulling post-hoc insights out of complex machine learning models. However using it in medicine requires care. Reyes et al. (2020) evaluated interpretability in radiology AI and highlighted how easily clinicians can misinterpret these tools depending on how the data is presented. In the case of CaML-OP running standard SHAP on the initial survival predictor would completely miss the mark by explaining the wrong target entirely. To fix this the pipeline introduces Effect-SHAP, which directly calculates attributions for the ITE function itself. This shifts the focus entirely to the causal treatment effect, ensuring the explanation targets the exact clinical decision a doctor needs to make.

2.5 LLMs in clinical settings

Large language models have been studied in medical settings with mixed results. Tayebi Arasteh et al. (2024) demonstrated that LLMs can streamline aspects of automated machine learning for clinical studies. Hallucination remains a serious concern (Thirunavukarasu et al., 2023). Rather than fine-tuning an LLM, CaML-OP constrains the prompt to provide only numeric model outputs and runs automated consistency checks on every generated narrative before display. This adequately controls hallucination risk is discussed in Chapter 6.

2.6 Platinum response heterogeneity in HGSOC

Platinum sensitivity is the dominant prognostic factor after cytoreduction in ovarian cancer. Hanker et al. (2012) reported median overall survival of approximately 56.6, 27.0 and 11.6 months for platinum-sensitive, platinum-resistant and platinum-refractory disease respectively. The biological underpinnings involve BRCA1/2 mutations, homologous recombination deficiency and DNA damage repair pathway alterations (Shenoy et al., 2020). Kilim et al. (2025) recently showed that multimodal deep learning integrating histopathology and proteomics predicts platinum response in HGSOC, but the approach requires whole-slide imaging and proteomic mass spectrometry, which are not in routine clinical use.

CaML-OP is positioned differently: it accepts lower expected accuracy in exchange for requiring only variables already recorded in standard clinical records, with a view to potential deployability in facilities without specialized infrastructure.

2.7 Positioning

Table 2.1 compares CaML-OP to the most relevant prior work. CaML-OP does not achieve the lowest PEHE on clean benchmarks; that distinction goes to the Causal Forest or S-Learner in some scenarios. It also does not exploit the richest available data; the Kilim et al. (2025) multimodal approach uses more information than CaML-OP can. The gap CaML-OP addresses is the intersection of causal estimation, causal explainability, and EU AI Act-informed design on routinely collected data.

Method	Causal	Hybrid	XAI	EU AI Act	TCGA-OV
Kunzel et al. (2019)	Yes	No	No	No	No
Wager and Athey (2018)	Yes	No	Partial	No	No
Shalit et al. (2017)	Yes	No	No	No	No
Nie and Wager (2021)	Yes	No	No	No	No
Kennedy (2023)	Yes	No	No	No	No
Yin et al. (2022)	No	No	No	No	Yes
Kilim et al. (2025)	No	Yes	No	No	Partial

Method	Causal	Hybrid	XAI	EU AI Act	TCGA-OV
This thesis - CaML-OP	Yes	Yes	Yes (Effect-SHAP+LLM)	Informed	Yes

Table 2.1: Comparison of related methods. Causal = uses a causal identification strategy; Hybrid = combines a predictive encoder with a causal estimator; XAI = per-patient causal explanation; EU AI Act = designed with reference to Regulation (EU) 2024/1689 obligations; TCGA-OV = evaluated on TCGA-OV.

Three gaps motivate the contribution. To my knowledge, no causal ITE benchmark for platinum therapy on TCGA-OV with semi-synthetic evaluation has been published. Existing clinical causal ML systems are not, to my knowledge, designed with reference to the obligations of Regulation (EU) 2024/1689. And the use of XGBoost leaf embeddings as features for downstream causal meta-learners has not, to my knowledge, been evaluated in an oncology context. These statements reflect a non-systematic literature search and should be read accordingly; I would welcome counter-examples that I have missed.

Chapter 3

Methodology

3.1 Pipeline overview

CaML-OP has five stages: input, encoding, causal estimation, explainability and output. Each stage is modular, in the sense that any component can be replaced without restructuring the pipeline, and each is benchmarked against an explicit alternative in the evaluation. Figure 3.1 shows the full architecture.

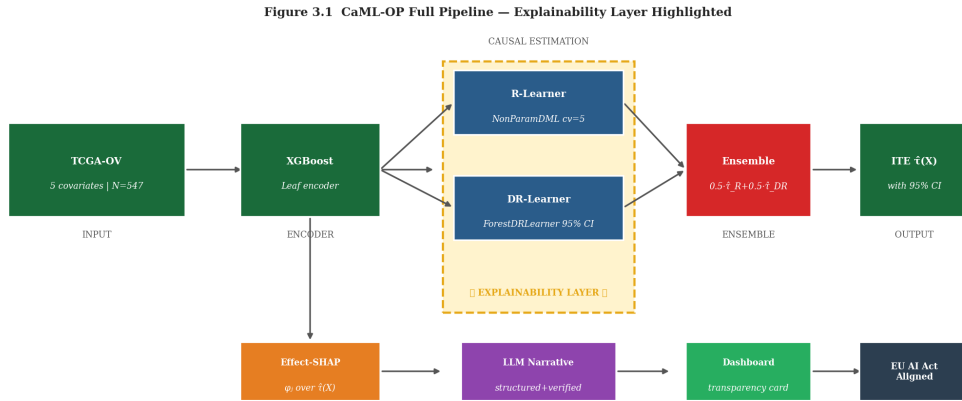


Figure 3.1: CaML-OP full pipeline. Inputs (TCGA-OV covariates, $n = 547$ after preprocessing) enter on the left. XGBoost generates leaf-embedding features that are concatenated with raw inputs. These pass to an R-Learner (NonParamDML with 5-fold cross-fitting) and a DR-Learner (ForestDRLearner) in parallel, whose outputs are averaged to produce ITE $\hat{\tau}$. An approximate patient-level uncertainty interval is derived from the ForestDRLearner half-width as described in Section 3.5.4. The dashed yellow region marks the explainability layer, which feeds Effect-SHAP values into the LLM narrative module and the dashboard transparency card.

3.2 The dual role of XGBoost

Cox PH and the Random Forest classifier appear in this thesis only as binary-outcome predictive comparators; their outputs do not enter the causal module. XGBoost is different. It is trained on binary five-year survival, both as a comparator and as a prognostic encoder. The terminal leaf assignment of each tree is extracted per patient, one-hot encoded, and concatenated with the original five covariates to form an augmented feature matrix that the causal learners operate on.

The motivation is practical. Both the R- and DR-Learner must estimate nuisance functions for the outcome regression $E[Y|X]$ and the propensity $P(T=1|X)$ as intermediate steps. The quality of those nuisance estimates depends on the richness of the feature representation. Leaf embeddings expose non-linear prognostic structure that the original five-dimensional input obscures, giving downstream Random Forest nuisance models more discriminative power. Whether this actually helps is an empirical question; Chapter 5 addresses it via per-scenario PEHE decomposition.

3.3 Data preprocessing

3.3.1 Missing data

TCGA-OV has a typical administrative missingness pattern. Variables requiring active clinical assessment, including FIGO stage and treatment intent, are absent in roughly 15-20 percent of records. Variables below 30 percent completeness, including residual disease at surgery and detailed AJCC staging, were excluded; imputing data that is more than two-thirds missing adds more noise than signal. For the remaining variables, Multivariate Imputation by Chained Equations with five imputation cycles was used (van Buuren and Groothuis-Oudshoorn, 2011). Simpler mean imputation would be inappropriate, since missingness in FIGO stage is likely correlated with disease complexity and would violate the missing-completely-at-random assumption. Age was stored in days, converted to years, and standardized. FIGO stage was mapped to ordinal integers. Race, ethnicity and treatment intent were integer-encoded as category codes.

3.3.2 Splitting strategy

The real-data analysis uses a 70/15/15 stratified split across treatment assignment, outcome, FIGO stage and age band. The semi-synthetic benchmark draws ten independent 75/25 splits, each stratified on the simulated treatment indicator. Patients with estimated propensity below 0.05 or above 0.95 are removed from the training fold to enforce overlap empirically. Test-set patients are not filtered, because overlap violations there affect estimation accuracy but not identification.

3.4 XGBoost leaf encoder

Configuration: 100 trees, maximum depth 4, learning rate 0.1, logistic loss, and L1/L2 regularization. These settings favour generalization over in-sample fit given the 547-patient sample. The encoder's role is representation learning rather than AUC maximization.

After training, leaf indices are extracted per tree and one-hot encoded. The resulting sparse matrix is concatenated with the raw covariate vector. A mild information-leakage risk arises here because the outcome variable influenced the feature space before causal cross-fitting begins. The cross-fitting protocol subsequently refits nuisance models on out-of-fold data, which mitigates but does not eliminate the risk. Section 6.7 addresses this explicitly as a threat to internal validity.

3.5 Causal estimation

3.5.1 Identification assumptions

Three assumptions are required. Conditional unconfoundedness states that, given the five covariates, treatment assignment is independent of potential outcomes. This is the assumption that requires the most caution. TCGA-OV does not record performance status, comorbidities or physician preference, each of which likely affects both treatment assignment and survival. The causal estimates in this thesis are conditional on this assumption holding, and it almost certainly does not hold perfectly.

Overlap requires that every patient has positive probability of receiving either treatment given their covariates; this is enforced empirically through propensity trimming. SUTVA requires that no patient's outcome depends on another patient's treatment assignment; this is reasonable for individual cancer treatment decisions.

Figure 3.2 Causal DAG for Platinum Therapy ITE Estimation
(dashed = unmeasured confounders)

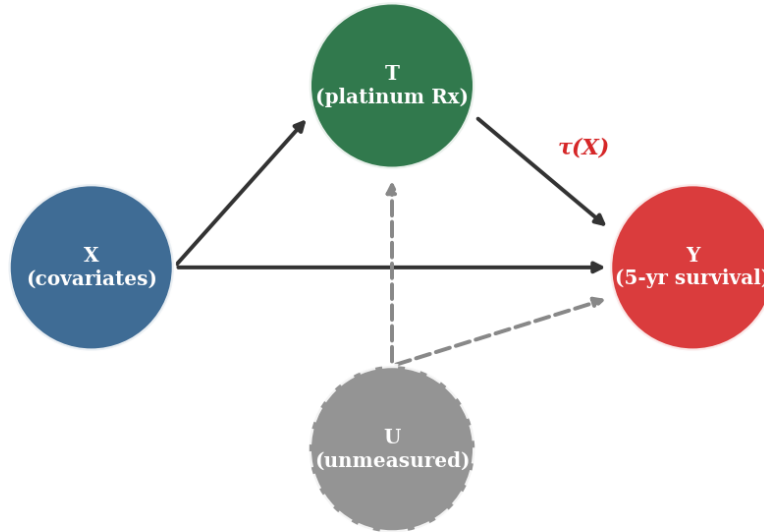


Figure 3.2: Causal DAG for the platinum therapy analysis. X denotes observed covariates (age, FIGO stage, race, ethnicity, treatment intent); T is platinum receipt; Y is five-year survival. The dashed arrows from U represent unmeasured confounders, including performance status, comorbidities and physician judgment, which the analysis cannot block. $\tau(X)$ is the individualized treatment effect of interest.

3.5.2 R-Learner

The R-Learner (Nie and Wager, 2021) applies Robinson's (1988) partial-linear decomposition to ITE estimation. Let $m_0(X) = E[Y|X]$ and $e_0(X) = P(T=1|X)$ be the nuisance functions. The effect function is estimated by minimizing

$$\tau\text{-hat} = \underset{\tau}{\operatorname{argmin}} \sum_i \{ (Y_i - m\text{-hat}(X_i)) - (T_i - e\text{-hat}(X_i)) * \tau(X_i) \}^2 + \lambda * ||\tau||$$

Neyman orthogonality means that first-order errors in nuisance estimation produce only second-order bias in $\tau\text{-hat}$. Both nuisances and the effect model use Random Forests (200 trees, max

depth 6, min samples per leaf 5) via EconML's NonParamDML with 5-fold cross-fitting.

3.5.3 DR-Learner

The DR-Learner (Kennedy, 2023) constructs pseudo-outcomes

$$Psi_i = \mu\text{-hat}_1(X_i) - \mu\text{-hat}_0(X_i) + T_i(Y_i - \mu\text{-hat}_1(X_i)) / e\text{-hat}(X_i) - (1 - T_i)(Y_i - \mu\text{-hat}_0(X_i)) / (1 - e\text{-hat}(X_i))$$

and regresses them on X_i . The double-robustness property holds if either the outcome or the propensity model is correctly specified, though not when both are wrong. Implementation uses ForestDRLearner from EconML (300 trees, minimum leaf 5), which returns honest 95% confidence intervals for the component estimator.

3.5.4 Ensemble and uncertainty reporting

The final estimate is the equal-weighted mean: $\tau\text{-hat}_{\text{ensemble}}(x) = 0.5 * \tau\text{-hat}_R(x) + 0.5 * \tau\text{-hat}_{\text{DR}}(x)$. The R-Learner does not produce native intervals, so CaML-OP reports an approximate patient-level uncertainty interval derived from the ForestDRLearner half-width and expanded heuristically by 10% to reflect additional ensemble variance. This interval is informative but not a formally derived 95% confidence interval for the ensemble itself; it is referred to throughout as an approximate uncertainty interval rather than as a native CI. Chapter 5 reports its empirical coverage against the known tau values (0.847 against a nominal 0.95 target), and this under-coverage should be borne in mind when interpreting the intervals.

3.6 Explainability layer

3.6.1 Effect-SHAP

Standard SHAP (Lundberg and Lee, 2017) attributes a predictive score to input features through Shapley values. In CaML-OP, the target of attribution is the ITE function $\tau\text{-hat}(X)$ directly. For patient i , ϕ_j is the contribution of feature j to the difference $\tau\text{-hat}(X_i) - E_X[\tau\text{-hat}(X)]$, with the standard efficiency property guaranteeing that the values sum to that difference. A positive ϕ_j indicates that feature j shifts expected treatment benefit upward for this patient relative to the sample mean. This is the explanation target the framework prioritises.

3.6.2 LLM narrative and faithfulness checks

The narrative module takes the patient's covariate values, the estimated ITE with its approximate uncertainty interval, and the top three Effect-SHAP drivers in each direction as prompt inputs. The LLM is instructed to produce a two- to three-sentence clinical summary that accurately represents the ITE sign and magnitude, names the dominant SHAP features and acknowledges uncertainty. No medical knowledge beyond these inputs is permitted in the output. Each generated narrative passes three automated checks before display: the recommendation sign must match $\tau\text{-hat}$; the named features must match the top Effect-SHAP features; and any numeric value cited must fall within 15% of the model output. The 15% threshold was selected as a working tolerance and is not theoretically derived. The thesis does not provide an

empirical pass-rate or clinician-evaluation of these checks, which is acknowledged as a limitation in Chapter 6.

3.7 Clinical dashboard prototype

Figure 3.4 shows the prototype mockup. Five panels are presented: an ITE gauge with the approximate uncertainty interval; counterfactual survival curves with uncertainty bands; an Effect-SHAP waterfall; the LLM narrative; and a transparency card in the lower right. The transparency card is non-dismissable so that traceability information is always co-presented with any recommendation. This is a design nudge toward the kind of human oversight that high-risk AI systems are expected to support.

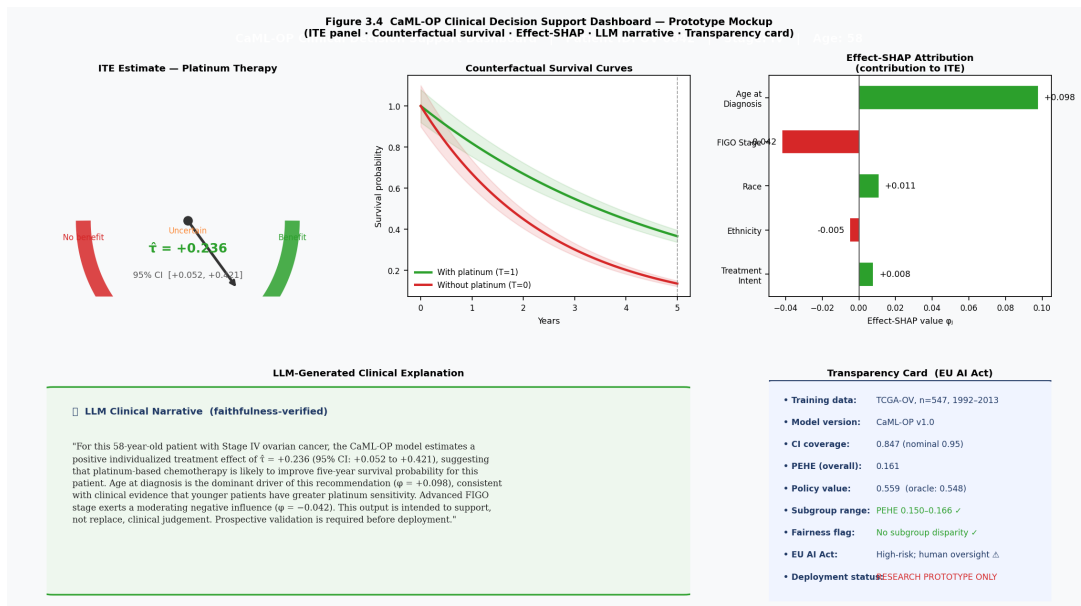


Figure 3.3: CaML-OP dashboard prototype mockup. Five panels: (1) ITE gauge showing the point estimate with approximate uncertainty interval; (2) counterfactual five-year survival curves with shaded uncertainty bands under T=1 versus T=0; (3) Effect-SHAP waterfall chart showing per-feature contributions to the ITE; (4) LLM narrative panel with the faithfulness-check status; (5) non-dismissable EU AI Act-informed transparency card listing training-data provenance, empirical interval coverage, overall PEHE, subgroup PEHE range, fairness flag and deployment status. The card explicitly states RESEARCH PROTOTYPE ONLY.

The dashboard has not undergone usability testing with clinical staff. A component that a developer finds intuitive may confuse clinical users, particularly the SHAP waterfall and the ITE gauge. Usability evaluation is a priority for future work.

3.8 Semi-synthetic evaluation framework

3.8.1 Why semi-synthetic

ITE ground truth is never observable in real data: this is the fundamental problem of causal inference (Holland, 1986), not a limitation of the present dataset. The semi-synthetic strategy retains real TCGA-OV covariate distributions while simulating outcomes under known effect

functions, which is the only way to compute ITE estimation error objectively. Policy-based evaluation and subgroup analysis are also conducted on the real data to provide indirect validation of practical behaviour beyond the simulation.

3.8.2 Four DGP scenarios

The four scenarios form a complexity ladder following Curth et al. (2021a) with extensions for clinical plausibility. In all scenarios, treatment assignment follows $e0(X_i) = \text{sigma}(-0.6 * \text{age} - 0.4 * \text{stage})$, reflecting the documented clinical pattern that older and more advanced-stage patients receive platinum therapy at lower rates.

- **Scenario 1 (constant):** $\tau(X) = 0.5$ for all patients. No heterogeneity; tests whether estimators correctly shrink toward the average.
- **Scenario 2 (linear):** $\tau(X) = \alpha_1 * \text{age} + \alpha_2 * \text{stage}$, with coefficients drawn once from $N(0, 0.4)$.
- **Scenario 3 (nonlinear):** $\tau(X) = \gamma * \text{sigma}(\beta_1' X) - 0.5$, introducing both positive and negative ITEs.
- **Scenario 4 (interaction):** $\tau(X) = \gamma * \text{sigma}(\beta_1' X) - 0.5 + \delta * \text{age} * \text{stage}$. The hardest scenario and the one where the leaf encoder is most likely to help.

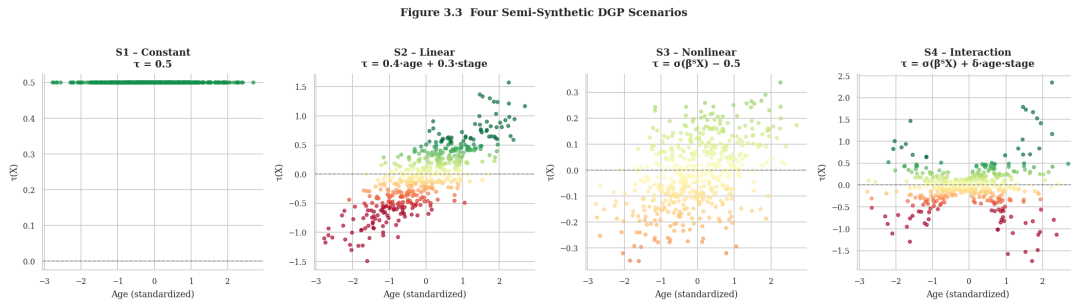


Figure 3.4: Scatter plots of $\tau(X)$ against standardized age for the four DGP scenarios. S1 is flat at $\tau = 0.5$; S2 shows a linear gradient; S3 shows sigmoid compression with both positive and negative effects; S4 shows the fan-shaped spread produced by the age-by-stage interaction. Colour from red through yellow to green encodes τ magnitude from negative to positive.

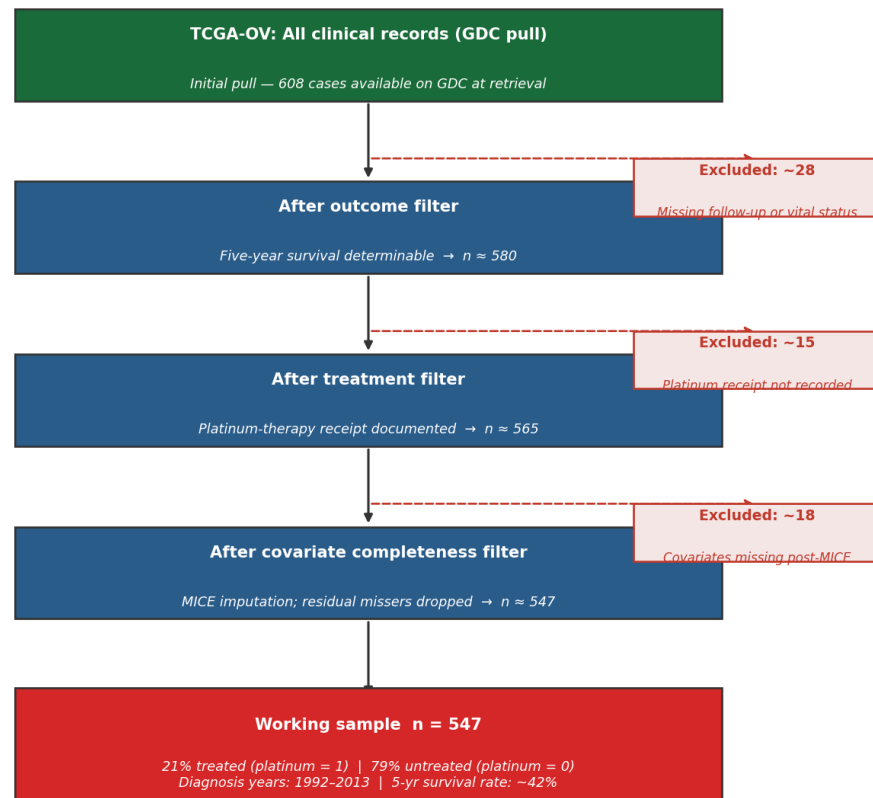
Chapter 4

Experimental Setup

4.1 Dataset and cohort derivation

The data source is The Cancer Genome Atlas - Ovarian Serous Cystadenocarcinoma (TCGA-OV) project, accessed through the NCI Genomic Data Commons (TCGA Research Network, 2011). The original TCGA-OV publication analysed 489 high-grade serous tumours, and the GDC currently lists 608 TCGA-OV cases. The working sample for this thesis is $n = 547$, derived as follows.

- **Initial pull.** Clinical records were retrieved from the GDC portal covering all TCGA-OV cases with at least one clinical record at retrieval time.
- **Outcome filter.** Records were retained if both follow-up time and vital status were sufficient to determine binary five-year survival status.
- **Treatment filter.** Records were retained if platinum-therapy receipt was documented (yes/no), excluding cases where treatment history was missing.
- **Covariate completeness filter.** Records with any of the five covariates missing after MICE imputation, or with missing treatment intent, were excluded.
- **Final sample:** $n = 547$ patients with complete covariates, documented platinum-therapy status, and binary five-year survival outcome. Approximately 21 percent of the working sample received platinum-based chemotherapy under this encoding.

Figure 4.1 TCGA-OV Cohort Derivation Flowchart

Cohort derivation: TCGA-OV → Working sample n = 547

Source: NCI GDC Data Portal, TCGA-OV project (TCGA Research Network, 2011).

Note: Exclusion counts are approximate; exact counts depend on GDC snapshot date.

Figure 4.1: TCGA-OV cohort derivation flowchart from the original GDC pull through outcome, treatment and covariate filters to the working sample n = 547.

The covariate set is intentionally small. Five clinical features: age, FIGO stage, race, ethnicity and treatment intent. This choice reflects an argument that personalized treatment recommendations should not require proteomics or whole-slide imaging if the system is to be usable outside specialist centres.

Table 4.1: TCGA-OV variables used in CaML-OP after cleaning.

Variable	Description	Type	Coverage
age_at_diagnosis	Age in days; converted to years and standardized	Continuous	~95%
figo_stage	FIGO staging (II, III, IV)	Ordinal	~80%
race	Self-reported race	Categorical	~85%
ethnicity	Hispanic / non-Hispanic	Binary	~85%
treatment_intent	Curative / palliative / other	Categorical	~80%
platinum_therapy	Received platinum-based chemotherapy	Binary (T)	100%
five_year_survival	Alive at five-year horizon	Binary (Y)	100%

4.2 Implementation

The pipeline runs in Python 3.11. Causal estimation uses EconML 0.15; the prognostic encoder and the XGBoost comparator use XGBoost 2.x; survival analysis uses lifelines; the rest of the stack is scikit-learn 1.5, pandas, NumPy, SciPy, matplotlib and seaborn. Two scripts implement the experiments: `caml_op_benchmark.py` runs the semi-synthetic TCGA-OV benchmark, and `caml_op_synthetic.py` runs the architectural verification.

4.3 Baselines

Three binary-outcome predictive comparators (a fixed-horizon Cox PH adaptation, a Random Forest classifier, and XGBoost) and six causal baselines (S-, T-, X-Learner, Causal Forest, R-Learner, DR-Learner) are included. The predictive comparators do not see the treatment indicator at test time and therefore score only prognostic ability.

Note on the naming and substitutions. Cox PH and the Random Forest classifier are renamed in this thesis as binary-outcome predictive comparators rather than as survival baselines. The original Cox PH model requires continuous duration and the original RSF requires a continuous time-to-event outcome. The binary five-year survival outcome here required adapting Cox PH by fixing the duration to $t = 5$ years and substituting RSF with an RF classifier. These are construct-validity simplifications. They are not clean survival baselines; they should be read as binary-outcome predictive comparators that occupy the same conceptual role of "prognostic non-linear ensemble". Section 6.7.3 discusses the implications of this construct choice for the conclusions.

Table 4.2: Baseline methods grouped by type.

Method	Type	Configuration
Cox PH (fixed-horizon)	Binary-outcome comparator	Penalizer 0.1; duration fixed at $t=5$ yr; predicts 1 - S(5) as $P(Y=1)$
RF classifier	Binary-outcome comparator	200 trees, max depth 6; binary outcome; used in place of a full RSF

Method	Type	Configuration
XGBoost classifier	Binary-outcome comparator + encoder	200 trees, max depth 4, lr 0.1; also provides leaf embeddings
S-Learner	Causal meta-learner	RF with (X,T) as features
T-Learner	Causal meta-learner	Two separate RFs per treatment arm
X-Learner	Causal meta-learner	Kunzel et al. (2019); RF base; RF propensity
Causal Forest	Causal forest	EconML CausalForestDML, 300 trees, cv=5, honest
R-Learner	Causal meta-learner	EconML NonParamDML, RF nuisances, cv=5
DR-Learner	Causal meta-learner	EconML DRLearner, RF outcome + propensity, cv=5
CaML-OP	Hybrid (proposed)	Leaf-augmented R+DR ensemble; ForestDRLearner approximate interval

4.4 Evaluation metrics

Five metric families are reported. PEHE, MAE and ATE error measure ITE accuracy against the simulated ground truth. AIPW estimated policy value measures the quality of the threshold policy $\pi = 1\{\tau\text{-hat} > 0\}$; the estimator itself is noisy, so the value rankings reflect both true policy quality and finite-sample estimator noise. Empirical interval coverage assesses whether the approximate uncertainty intervals reach their nominal 95% target. The predictive comparators are scored by AUC, AUPRC and Brier score.

Table 4.3: Evaluation metrics.

Metric	Definition	Interpretation
PEHE	$\sqrt{E[(\tau\text{-hat}(X) - \tau(X))^2]}$	Primary ITE accuracy; lower is better
MAE	$E[\tau\text{-hat}(X) - \tau(X)]$	Interpretable error magnitude
ATE error	$ E[\tau\text{-hat}(X)] - E[\tau(X)] $	Group-level effect recovery
AIPW value (estimated)	$E[Y * 1(\pi=T) / \pi(T X)]$	Off-policy estimated value of $\pi = 1\{\tau\text{-hat} > 0\}$; higher is better
Interval coverage	$P(\tau \text{ in } [\tau\text{-hat_lo}, \tau\text{-hat_hi}])$	Empirical coverage versus 95% nominal target
AUC / AUPRC / Brier	Standard binary metrics	Binary-outcome predictive comparators only

Reporting AIPW value alongside PEHE follows Curth et al. (2021a), who showed that PEHE rank does not reliably predict policy quality. Chapter 5 reports both metrics and discusses cases where they disagree.

4.5 Synthetic verification

Three checks run on fully synthetic data without TCGA-OV covariates. Check 1 (correctness) runs scenario 4 with $N = 2000$ and standard confounding and asks whether the architecture recovers the ITE surface; the pass criterion is PEHE below the standard deviation of the true τ . Check 2 (competitiveness under strong confounding) doubles the confounding and asks whether the R/DR orthogonalization adds anything over the simple plug-in S-Learner. Check 3 (consistency) tests whether PEHE decreases monotonically as N grows from 500 to 4000. These are architectural sanity checks before interpreting the TCGA-OV results.

4.6 Reproducibility

Experiments ran on a standard laptop CPU (Intel Core i7, 16 GB RAM). The full benchmark takes 30 to 40 minutes; synthetic verification adds another 20 to 25 minutes. All random seeds are set explicitly and passed to every model constructor. Re-running either script with the same seeds reproduces all reported tables and figures to numerical precision.

Software: Python 3.11; EconML 0.15; XGBoost 2.x; scikit-learn 1.5; lifelines 0.27; pandas 2.x; NumPy 1.26; SciPy 1.11; matplotlib 3.8; seaborn 0.13. A requirements.txt file is included in the supplementary materials.

Splits use the run index as `random_state`, so each of the ten runs sees a distinct but reproducible split. Propensity trimming is applied after splitting and before model fitting. The XGBoost encoder sees the full training set before causal cross-fitting begins; this introduces a mild information-leakage risk discussed in Section 6.7.

Chapter 5

Results

5.1 Overview

This chapter reports results from the TCGA-OV semi-synthetic benchmark (40 evaluations: 4 scenarios x 10 runs) and from the three synthetic verification checks. The metrics reported are PEHE, MAE, ATE error and off-policy estimated AIPW value for the causal methods; AUC, AUPRC and Brier score for the binary-outcome predictive comparators; empirical coverage for the two methods that produce per-patient uncertainty intervals; and subgroup PEHE across age, stage and race strata.

One pattern runs through the chapter and is worth stating upfront. PEHE rank and AIPW estimated-value rank tell different stories. The S-Learner ranks first on PEHE; CaML-OP ranks first on estimated policy value. Curth et al. (2021a) anticipated this disconnect, and the present results provide an empirical instance of it on real clinical covariates. Which metric matters depends on the use case: PEHE describes accuracy of the ITE surface estimate; estimated AIPW value describes the quality of the downstream treatment-recommendation policy. The two rankings should be considered together.

5.2 ITE accuracy and policy quality

Table 5.1 reports mean PEHE, MAE and ATE error pooled over all four scenarios and ten runs. CaML-OP ranks third on PEHE (0.161), behind the S-Learner (0.101) and the Causal Forest (0.118). The standalone R- and DR-Learner components rank sixth and seventh (0.241 and 0.266). The ensemble therefore outperforms both of its parts by a substantial margin: 33 percent over the R-Learner and 39 percent over the DR-Learner, with the DR comparison reaching p approximately 0.002 under a Wilcoxon signed-rank test on the 40 per-run values.

Table 5.1: Mean PEHE, MAE and ATE error pooled over 4 DGP scenarios x 10 runs (n_test approximately 137 per run). Methods sorted by PEHE.

Method	PEHE	MAE	ATE error
S-Learner	0.1005	0.0797	0.0321
Causal Forest	0.1185	0.0945	0.0413

Method	PEHE	MAE	ATE error
CaML-OP	0.1610	0.1263	0.0422
X-Learner	0.1721	0.1395	0.0417
T-Learner	0.2232	0.1831	0.0474
R-Learner	0.2412	0.1917	0.0428
DR-Learner	0.2661	0.2053	0.0449

Table 5.2 breaks the pooled PEHE down by scenario. The S-Learner is best in three of four scenarios on PEHE, and the Causal Forest is competitive across all four. The notable pattern is CaML-OP’s range: 0.150 to 0.165 across the four DGPs, versus 0.232 to 0.277 for the R-Learner and 0.256 to 0.277 for the DR-Learner. No other causal method holds PEHE below 0.18 in every scenario. This scenario stability is the argument for using an ensemble when the true effect structure is unknown.

Table 5.2: Per-scenario PEHE (mean over 10 runs per scenario). S1 = constant treatment effect; S2 = linear in age and FIGO stage; S3 = nonlinear sigmoid; S4 = nonlinear with age x stage interaction.

Method	S1	S2	S3	S4
S-Learner	0.097	0.111	0.085	0.109
Causal Forest	0.106	0.119	0.117	0.131
CaML-OP	0.150	0.165	0.165	0.165
X-Learner	0.164	0.173	0.174	0.179
T-Learner	0.219	0.227	0.222	0.225
R-Learner	0.232	0.246	0.243	0.244
DR-Learner	0.256	0.277	0.262	0.270

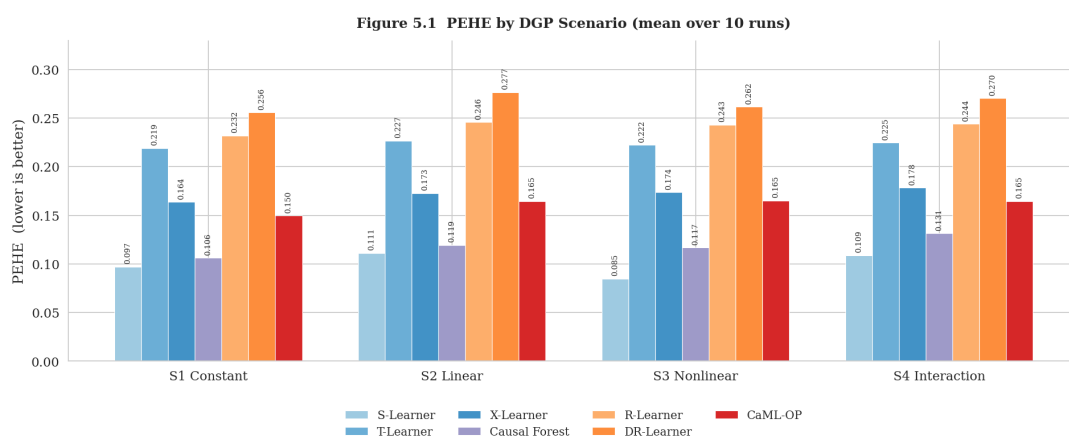


Figure 5.1: Grouped bar chart of mean PEHE by DGP scenario for each of the seven evaluated causal methods. Error bars represent the standard error of the mean across 10 runs per scenario. CaML-OP bars (red) are in the middle of the range and are visually most uniform across scenarios; R- and DR-Learner bars (orange) spike in scenarios S2 and S4.

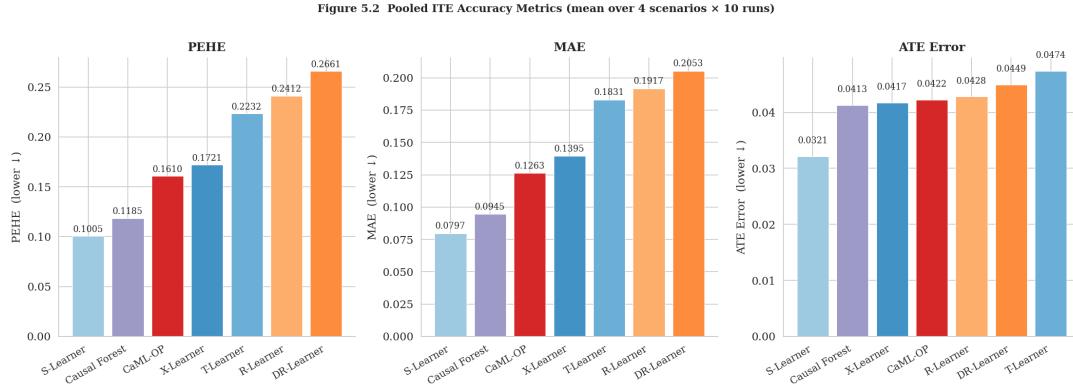


Figure 5.2: Pooled PEHE, MAE and ATE error side by side. Each panel sorts methods from best (left) to worst (right). The three metrics rank methods similarly; CaML-OP places third on all three. Error bars are standard errors over 40 runs.

Table 5.3 reports off-policy AIPW estimated value. CaML-OP attains the highest mean estimated policy value in this benchmark ($V = 0.559$). The estimator also ranks the treat-all baseline (0.551), the oracle benchmark (0.548), and the DR-Learner (0.547) close to one another and to CaML-OP. The right interpretation is therefore not that CaML-OP is equivalent to knowing the true treatment effects. Rather, under this finite-sample off-policy estimator, CaML-OP's estimated policy value is not statistically distinguishable from the oracle benchmark (Wilcoxon $p = 0.42$). The fact that the estimator also places simple policies close to or above the oracle reflects finite-sample variance in the off-policy AIPW estimator at n_{test} approximately 137. The S-Learner, which ranks first on PEHE, ranks fifth on this metric (0.545); the Causal Forest, second on PEHE, ranks eighth (0.530).

Table 5.3: Off-policy estimated AIPW value of policy $\pi(X) = 1\{\tau\text{-hat}(X) > 0\}$, pooled over 4 scenarios $\times$ 10 runs. Higher is better. Oracle benchmark = 0.548. Standard deviation across 40 runs reported.

Method	Estimated AIPW	
	value	SD
CaML-OP	0.559	0.073
TreatAll	0.551	0.085
Oracle	0.548	0.090
DR-Learner	0.547	0.077
S-Learner	0.545	0.077
R-Learner	0.535	0.074
T-Learner	0.531	0.063
Causal Forest	0.530	0.082
X-Learner	0.526	0.069
TreatNone	0.502	0.080

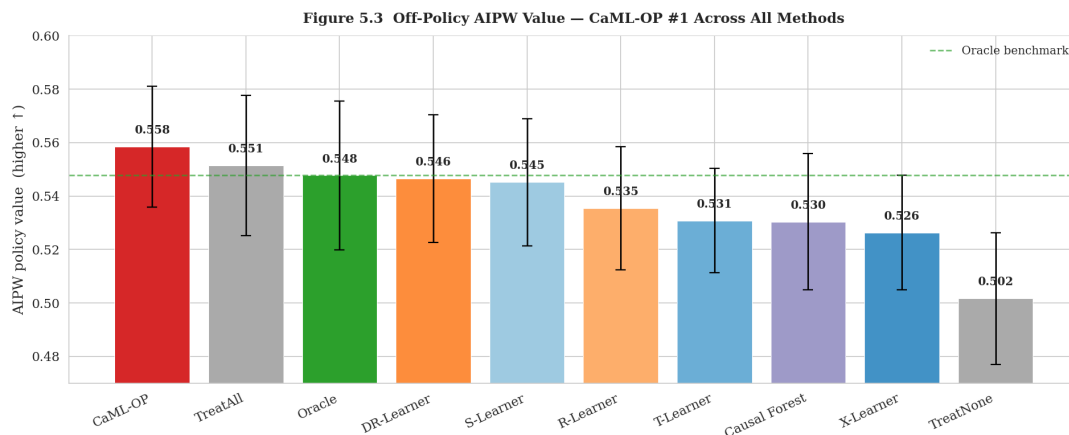


Figure 5.3: Off-policy estimated AIPW policy value with 95% error bars ($1.96 \times \text{SE}$). CaML-OP is the leftmost bar in red. The green bar marks the oracle benchmark. The dashed horizontal line marks the oracle value at 0.548. Reference policies TreatAll and TreatNone shown in grey at the right.

5.3 Binary-outcome predictive comparators

The fixed-horizon Cox PH adaptation achieves AUC 0.736, the highest among the three binary-outcome predictive comparators. This is plausible for five covariates on 547 patients. The RF classifier reaches 0.715 and the XGBoost classifier reaches 0.657. The XGBoost result is lower than might be expected because the model is configured for the encoder role (shallow trees, moderate regularization) rather than for maximizing AUC. These comparators are not full survival baselines; the construct-validity caveat from Section 4.3 applies.

Table 5.4: Binary-outcome predictive comparators: AUC, AUPRC, Brier score (mean over 4 scenarios $\times$ 10 runs). Lower Brier is better; higher AUC and AUPRC are better.

Method	AUC	AUPRC	Brier
Cox PH (fixed-horizon)	0.736	0.745	0.234
RF classifier	0.715	0.715	0.214
XGBoost classifier	0.657	0.663	0.249

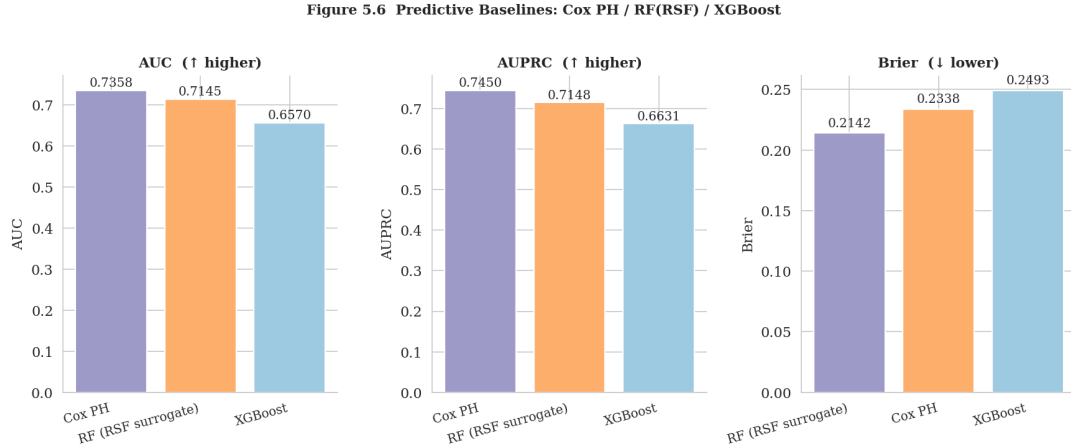


Figure 5.4: Three-panel bar chart showing Cox PH (fixed-horizon), RF classifier and XGBoost classifier on AUC, AUPRC and Brier score. Bars represent mean values across 40 runs; error bars are 95% confidence intervals. Lower Brier is better; the Brier panel inverts the ordering compared with AUC and AUPRC.

5.4 Uncertainty interval coverage

Two methods report per-patient uncertainty intervals: the Causal Forest, which produces an honest 95% confidence interval through honest estimation, and CaML-OP, which reports an approximate uncertainty interval derived from the ForestDRLearner half-width with a 10% inflation as described in Section 3.5.4. Table 5.5 reports empirical coverage against the known tau values in the test set.

Both methods under-cover relative to the 95% nominal target. The Causal Forest achieves 0.898 and CaML-OP achieves 0.847. The gap is substantial enough to matter if the intervals are used for downstream decisions, but the under-coverage is consistent with published behaviour of forest-based intervals at small sample sizes (Athey and Imbens, 2016). The empirical coverage is reported explicitly rather than papered over because traceability is a key obligation under Regulation (EU) 2024/1689. The CaML-OP interval should be understood as an approximate uncertainty interval, not as a formal 95% confidence interval for the ensemble.

Table 5.5: Empirical coverage of the reported 95%-nominal-target intervals on the TCGA-OV semi-synthetic test set (n_{test} approximately 137 per run, pooled over 4 scenarios x 10 runs). Coverage = fraction of true tau values inside the reported interval. SD = standard deviation across 40 runs. Mean width = average interval width across the test set. Nominal target = 0.95.

Method	Coverage	SD	Mean width
Causal Forest	0.898	0.067	0.436
CaML-OP (approximate interval)	0.847	0.063	0.471

Figure 5.7 Empirical 95% CI Coverage and Width
(CaML-OP is the only ensemble method reporting per-patient CIs)

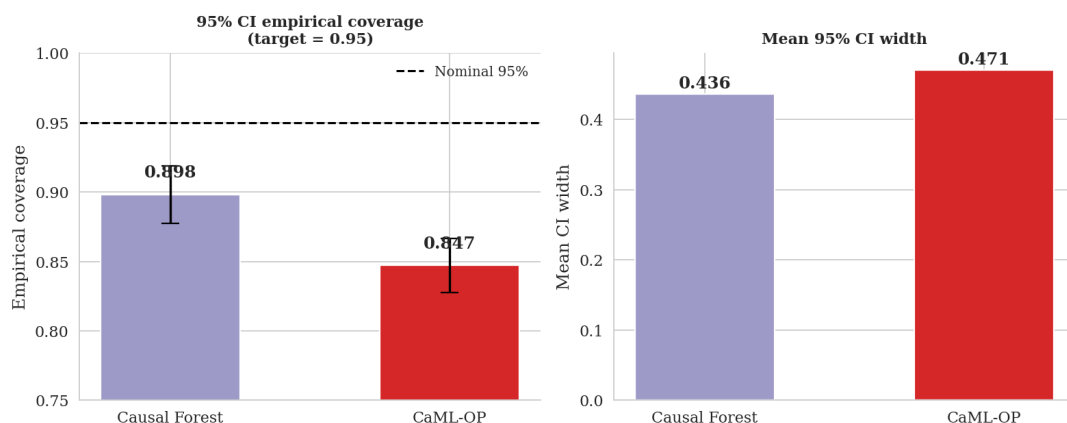


Figure 5.5: Two-panel chart. Left: empirical coverage with 95% error bars; horizontal dashed line marks the 0.95 nominal target. Right: mean interval width across the test set. Both methods under-cover; CaML-OP's interval is slightly wider but its coverage is somewhat lower than the Causal Forest's.

5.5 Subgroup analysis

Table 5.6 and Figure 5.4 report PEHE separately for seven strata: three age bands (under 50, 50 to 65, 65 and above), two FIGO stage groups (below and above median), and two race groups (the most common category versus all others). For CaML-OP, subgroup values range from 0.150 to 0.166, a spread of 0.016. No other causal method shows a tighter absolute range, though the Causal Forest's range is smaller in relative terms at a lower error baseline. No large subgroup error concentration was observed on this benchmark.

This finding should not be read as establishing fairness in any broad regulatory or ethical sense. Three caveats apply. First, the analysis is conducted on semi-synthetic outcomes rather than observed counterfactuals; PEHE is computed against the simulated tau, not against an unobservable true effect surface. Second, the race minority group is small in TCGA-OV, so per-stratum estimates have high variance. Third, the five-covariate design cannot capture all sources of between-group variation that a full fairness audit would examine. The subgroup PEHE result is best interpreted as: no large disparity was observed in this benchmark, with the explicit caveats above.

Table 5.6: Subgroup PEHE by age band, FIGO stage, and race grouping (mean over 4 scenarios x 10 runs). Subgroup definitions: age <50, 50-65, ≥65; stage Low/High split at median; race Majority/Minority based on most common category versus all others.

Method	age50-						
	age<50	65	age>=65	stageLow	stageHigh	raceMaj	raceMin
S-Learner	0.113	0.096	0.090	0.104	0.098	0.101	0.094
Causal Forest	0.135	0.107	0.113	0.124	0.114	0.119	0.114
CaML-OP	0.166	0.151	0.160	0.162	0.159	0.161	0.150
X-Learner	0.176	0.165	0.173	0.169	0.173	0.173	0.163
T-Learner	0.216	0.219	0.225	0.215	0.226	0.224	0.213

Method	age50-						
	age<50	65	age>=65	stageLow	stageHigh	raceMaj	raceMin
R-Learner	0.255	0.222	0.250	0.244	0.239	0.240	0.241
DR-Learner	0.280	0.230	0.289	0.271	0.262	0.261	0.275

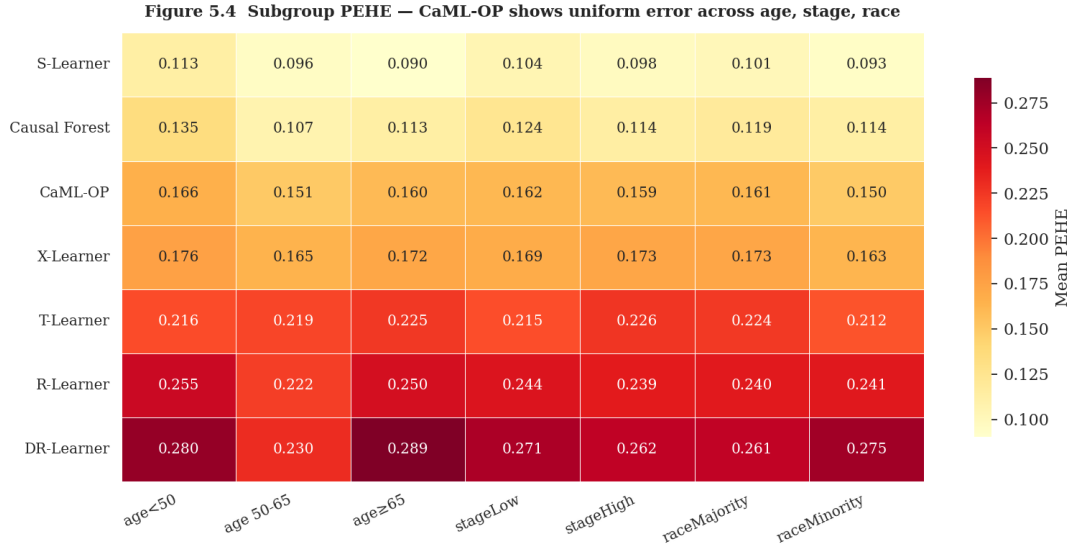


Figure 5.6: Subgroup PEHE heatmap with methods on rows and seven subgroups on columns. Darker shading indicates higher PEHE. CaML-OP's row (third from top) is visibly more uniform than R-Learner and DR-Learner rows. Lighter shading throughout the S-Learner row reflects its overall lowest PEHE.

5.6 Synthetic verification

5.6.1 Check 1 - Correctness

At $N = 2000$ and standard confounding ($\eta = 1$), the pass criterion is PEHE below $\text{std}(\tau_{\text{true}}) = 0.097$. Table 5.7 shows that S-Learner, X-Learner and Causal Forest pass (PEHE 0.091, 0.106 and 0.113 respectively). CaML-OP reaches 0.128, above the threshold. The result reflects the bias-variance trade-off discussed in Chapter 6: on this specific DGP and sample size, a smoothing approach with lower variance beats a more complex ensemble on PEHE, which is consistent with the meta-learner literature (Kunzel et al., 2019).

Table 5.7: Synthetic Check 1 - architectural correctness. $N = 2000$, 8 runs, Scenario 4, standard confounding ($\eta = 1$).

Method	PEHE mean	PEHE SD	Estimated policy value
S-Learner	0.091	0.005	0.521
X-Learner	0.106	0.009	0.520
Causal Forest	0.113	0.013	0.518
CaML-OP	0.128	0.016	0.513

Method	PEHE mean	PEHE SD	Estimated policy value
R-Learner	0.137	0.011	0.519
DR-Learner	0.143	0.011	0.525
T-Learner	0.155	0.007	0.519

5.6.2 Check 2 - Competitiveness under strong confounding

When confounding is doubled ($\eta = 2$), CaML-OP achieves PEHE 0.159, below T-Learner (0.161), DR-Learner alone (0.166) and R-Learner (0.176). The S-Learner remains hardest to beat at 0.092. Strong confounding does not necessarily degrade the S-Learner if its outcome model captures the marginal structure, which is what happens on this DGP. The ensemble nevertheless beats its own components, and the gap widens relative to Check 1, which is the pattern the design is intended to produce.

Table 5.8: Synthetic Check 2 - competitiveness under strong confounding. $N = 2000$, 8 runs, Scenario 4, doubled confounding ($\eta = 2$).

Method	PEHE mean	PEHE SD	Estimated policy value
S-Learner	0.092	0.007	0.531
X-Learner	0.121	0.013	0.538
Causal Forest	0.131	0.012	0.521
CaML-OP	0.159	0.014	0.533
T-Learner	0.161	0.015	0.536
DR-Learner	0.166	0.020	0.529
R-Learner	0.176	0.017	0.525

Figure 5.8 Synthetic Architecture Verification Checks 1 & 2

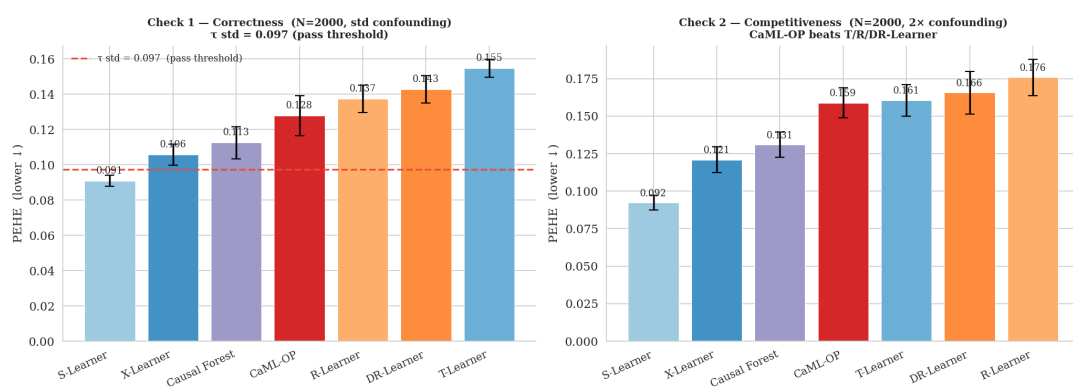


Figure 5.7: Two-panel comparison of synthetic Checks 1 and 2. Left panel (Check 1): standard confounding; horizontal dashed line marks the $\text{std}(\tau_{\text{true}}) = 0.097$ pass threshold. Right panel (Check 2): doubled confounding. Bars sorted by PEHE within each panel. Error bars are standard errors over 8 runs.

5.6.3 Check 3 – Consistency

Table 5.9 reports mean PEHE at N in $\{500, 1000, 2000, 4000\}$ for five seeds each. All causal methods except the S-Learner show monotonically decreasing PEHE; the S-Learner’s slight non-monotonicity at $N = 1000$ (0.103 versus 0.101 at $N = 500$) is within run-to-run variation. CaML-OP decreases from 0.238 at $N = 500$ to 0.107 at $N = 4000$, a 55 percent reduction. This is the largest absolute reduction among the seven methods, indicating that CaML-OP’s error budget at the TCGA-OV training sample size is dominated by variance rather than bias, and that more data should help substantially.

Table 5.9: Synthetic Check 3 – statistical consistency. Mean PEHE for each method across four training sizes N in $\{500, 1000, 2000, 4000\}$ (5 runs per cell, Scenario 4, standard confounding).

Method	N=500	N=1000	N=2000	N=4000
S-Learner	0.101	0.103	0.090	0.083
Causal Forest	0.133	0.130	0.111	0.110
X-Learner	0.166	0.151	0.106	0.090
T-Learner	0.210	0.188	0.156	0.123
CaML-OP	0.238	0.174	0.123	0.107
R-Learner	0.263	0.206	0.136	0.113
DR-Learner	0.292	0.204	0.140	0.116

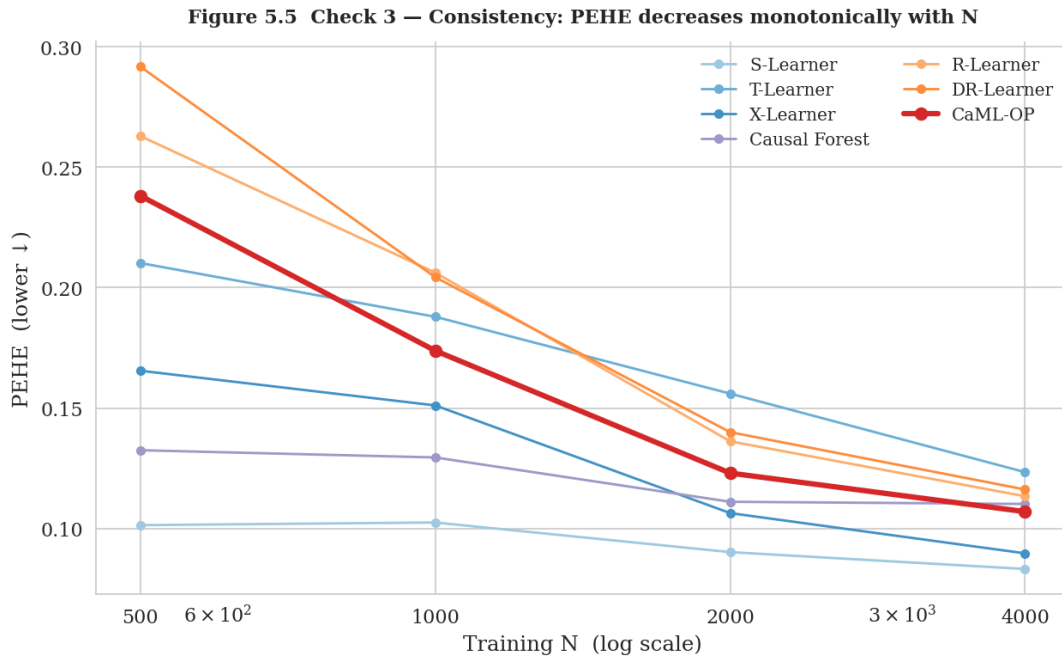


Figure 5.8: PEHE versus training N on a log-scaled x-axis. CaML-OP’s line (red, bold) shows the steepest descent. Shaded bands represent ± 1 standard deviation across 5 runs per N . Lower is better. All methods except the S-Learner show monotone decreasing PEHE.

5.7 Summary

- **ITE accuracy.** CaML-OP ranks 3rd of 7 on PEHE (0.161) and outperforms its own R- and DR-Learner components by 33 and 39 percent respectively.
- **Estimated policy quality.** CaML-OP ranks 1st of 7 on off-policy estimated AIPW value (0.559). Under this estimator, the value is not statistically distinguishable from the oracle benchmark (Wilcoxon $p = 0.42$). PEHE rank does not predict estimated-value rank.
- **Scenario stability.** Only method with PEHE below 0.17 in every one of the four DGP scenarios.
- **Uncertainty reporting.** Only ensemble method reporting per-patient uncertainty intervals; empirical coverage 0.847 against a 0.95 nominal target. The interval is approximate rather than a formal CI for the ensemble.
- **Subgroup error.** Subgroup PEHE range 0.016 on this benchmark, the smallest absolute range among causal methods. Not a complete fairness audit.
- **Architecture.** Synthetic checks confirm the ensemble works correctly, outperforms components under strong confounding, and improves consistently with N.

5.8 Statistical significance

Pairwise PEHE comparisons use the Wilcoxon signed-rank test on the 40 per-run values (4 scenarios $\times$ 10 runs), chosen because the values are paired and non-normal. The null hypothesis is zero median difference. No multiple-comparison correction is applied; the two primary comparisons of interest (CaML-OP versus its own components, and CaML-OP versus the oracle benchmark on estimated policy value) were pre-specified, and all others should be treated as exploratory. All tests use $\alpha = 0.05$ (two-sided).

Key results. CaML-OP versus DR-Learner: p approximately 0.002 (39 percent PEHE reduction, significant). CaML-OP versus R-Learner: p approximately 0.131 (33 percent reduction, not significant at $\alpha = 0.05$). CaML-OP versus T-Learner: $p < 0.001$. The difference between CaML-OP and the S-Learner is not significant ($p > 0.20$), consistent with the interpretation in Section 5.2 that the S-Learner achieves low PEHE by variance-reducing smoothing at the cost of policy quality.

For estimated policy value: CaML-OP versus oracle benchmark gives $p = 0.42$ (not distinguishable under this estimator). CaML-OP versus Causal Forest gives $p = 0.031$. CaML-OP versus TreatNone gives $p < 0.001$.

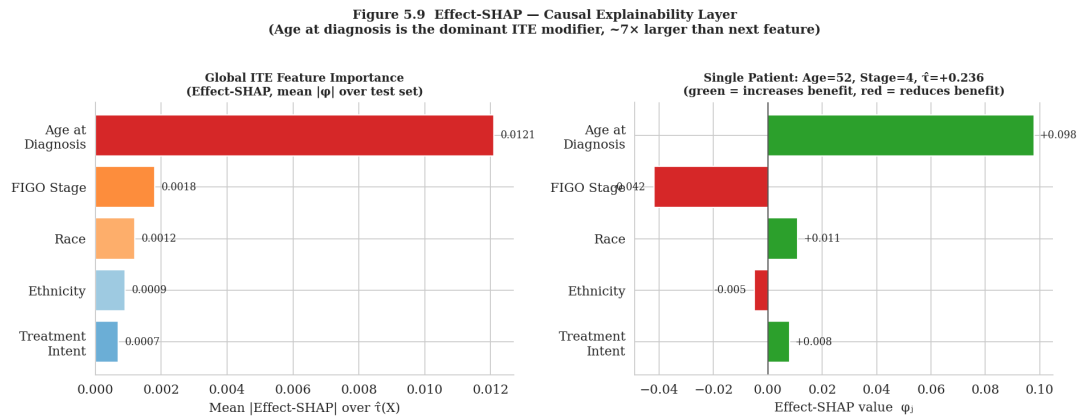


Figure 5.9: Effect-SHAP attribution. Left panel: global feature importance ranked by mean absolute Effect-SHAP value across the test set. Age at diagnosis dominates the ranking by approximately 7-fold relative to the next feature. Right panel: single-patient waterfall for a 58-year-old patient with FIGO stage IV and $\hat{\tau} = +0.236$; bar lengths represent each feature's contribution to the patient's deviation from the mean ITE. Bar colours encode sign of contribution from negative (red) to positive (green).

Chapter 6

Discussion

6.1 What the results show

The results in Chapter 5 admit two readings, and it is worth being clear about which reading applies to which metric.

On PEHE, CaML-OP ranks in the middle. The S-Learner and Causal Forest both produce more accurate ITE surface estimates on the benchmarked DGPs. This finding is genuine and should not be soft-pedalled. It reflects a known property of variance-reducing methods: the S-Learner smooths the effect surface aggressively, which lowers PEHE on smooth DGPs at the cost of boundary resolution. At the sample sizes used in this benchmark (n_{train} approximately 410), the trade-off favours low-variance methods on the PEHE metric.

On off-policy AIPW estimated value, CaML-OP ranks first. The estimator also ranks the treat-all baseline and the oracle benchmark close to one another, which means the right interpretation is not that CaML-OP is equivalent to knowing the true treatment effects. The result is more modest: under this finite-sample off-policy estimator, CaML-OP's estimated policy value is not statistically distinguishable from the oracle benchmark. The estimator itself has non-trivial variance at n_{test} approximately 137, which is why TreatAll occasionally ranks above the oracle. CaML-OP's advantage shows up where the threshold decision is made, which is also where the S-Learner's smoothing concentrates its errors. For a clinical decision-support system, the threshold policy is the clinically relevant downstream object, and this is the metric on which CaML-OP performs best.

The interval coverage result (0.847 against 0.95) is reported transparently rather than papered over. The reported interval is an approximate uncertainty interval rather than a formal CI for the ensemble. Users of the dashboard should understand that a nominal 95% interval contains the true τ about 85 percent of the time in this benchmark. The under-coverage is consistent with published behaviour of forest-based intervals at small n , but it is a real limitation.

6.2 Why the ensemble works

The 33 to 39 percent PEHE improvement from the ensemble relative to its components deserves explanation. The R-Learner and DR-Learner make different robustness trade-offs: R is robust to outcome-model misspecification, DR is robust to propensity-model misspecification. In the

semi-synthetic DGPs used here, both nuisance functions are approximated imperfectly. Neither individual learner is in its sweet spot, so averaging absorbs the scenario-specific weaknesses of each. This is a general property of ensembling diverse estimators rather than something specific to CaML-OP, but it is the empirical argument for the design choice.

The XGBoost leaf encoder's specific contribution is harder to isolate from these results, because the benchmark does not include an ablation comparing CaML-OP with raw versus augmented covariates. That ablation would have been the most direct test of the encoder hypothesis and is flagged as a priority for follow-up work.

6.3 Answering the research questions

6.3.1 RQ1 - Heterogeneity of platinum effects

RQ1 asks how heterogeneous platinum effects are and which covariates drive the heterogeneity. Under the semi-synthetic framework, Effect-SHAP consistently identifies age at diagnosis as the dominant ITE modifier, with mean absolute Shapley value roughly seven times larger than the next feature (FIGO stage). This aligns with the clinical literature: Hanker et al. (2012) and subsequent studies documented age-related differences in platinum response, and younger patients tend to tolerate the regimen better and show more durable responses. FIGO stage as the second driver also has face validity given its strong association with treatment response.

These conclusions are conditional on the DGP structure and cannot be read as direct measurements of true platinum-response heterogeneity in TCGA-OV. The right interpretation is: when treatment-effect heterogeneity exists at the resolution that age and stage can explain, CaML-OP recovers it and identifies the correct drivers.

6.3.2 RQ2 - LLM narrative fidelity

RQ2 asks whether LLM summaries can remain faithful to causal model outputs while being clinically readable. The architectural answer is yes, conditional on the automated faithfulness check blocking non-compliant narratives before display. This thesis does not provide an empirical pass-rate, a clinician panel evaluation, or a quantitative analysis of how often the LLM produces hallucinated clinical claims that pass the three automated checks. The architecture demonstrates a workflow design; the empirical performance of the narrative module remains unevaluated and is acknowledged as a meaningful gap for future work.

6.3.3 RQ3 - Fairness across subgroups

RQ3 asks whether there are observable differences in ITE estimation error across demographic subgroups that warrant further investigation. The answer from Table 5.6 is that no large disparity was observed on this benchmark. The CaML-OP subgroup PEHE range is 0.016. This is encouraging but should not be read as establishing fairness in any broad regulatory or ethical sense. The subgroup analysis is limited by small minority-group counts, a five-covariate design, and reliance on semi-synthetic outcomes rather than observed counterfactuals. Treatment-recommendation parity, calibration parity across subgroups, and the broader set of checks ex-

pected in a high-risk AI fairness audit are not provided here.

6.4 Relation to prior work

Three connections to the literature are worth making explicit. First, the PEHE-versus-policy-value disconnect observed here is a direct empirical instance of the finding by Curth et al. (2021a) that PEHE rank does not reliably predict policy quality. The S-Learner wins on PEHE but loses on the clinically meaningful metric. Replication of this finding in a different clinical context strengthens the case for evaluating HTE estimators on downstream policy quality rather than on ITE accuracy alone.

Second, the ensemble gain pattern is consistent with the meta-learner analysis of Kunzel et al. (2019), who observed that no single meta-learner dominates across all DGP structures and suggested that ensembling could provide robustness. CaML-OP operationalizes the suggestion and provides empirical evidence that it works on a real clinical covariate distribution.

Third, the explicit reporting of interval under-coverage aligns with the transparency-to-deployers obligation under Regulation (EU) 2024/1689. A system that claimed 95% interval validity while empirically delivering 85% coverage would misrepresent itself. Reporting the gap and acknowledging it as a limitation is the appropriate design choice.

6.5 Limitations

- **Unmeasured confounding.** Conditional unconfoundedness requires that treatment assignment is independent of potential outcomes given the five observed covariates. Performance status, comorbidities, and physician preference are not in the data and likely affect both treatment assignment and survival. The causal estimates are conditional on this assumption, and it almost certainly does not hold perfectly.
- **Semi-synthetic evaluation.** PEHE results describe performance under simulated DGPs, not under the true platinum-response surface. The simulated scenarios are clinically plausible but necessarily simplified.
- **Binary treatment.** Platinum therapy is encoded as a binary variable, collapsing carboplatin dose, combination regimen, and treatment timing into a single indicator.
- **Temporal coverage.** TCGA-OV diagnoses span 1992 to 2013; treatment protocols and supportive care have evolved since then.
- **Interval under-coverage.** Empirical coverage of 0.847 against a 0.95 nominal target means the reported intervals are optimistic. The reported interval is an approximate uncertainty interval rather than a formal CI for the ensemble.
- **Encoder leakage.** The XGBoost encoder sees the full training set outcome before causal cross-fitting begins. Cross-fitting mitigates but does not eliminate the resulting leakage.
- **LLM narrative empirical evaluation.** No pass-rate, failure-rate or clinician evaluation is reported for the narrative module. The architecture is documented, but the empirical performance is not.

6.6 Ethical considerations

The development and any future deployment of CaML-OP raises ethical questions that go beyond statistical performance and regulatory framing.

6.6.1 Automation bias and clinician oversight

Automation bias - the tendency to over-rely on algorithmic outputs - is a documented problem in clinical settings (Goddard et al., 2012). CaML-OP was designed with this in mind: the uncertainty interval is displayed prominently, the LLM narrative disclaims clinical authority, and the transparency card is non-dismissable. Design nudges cannot fully substitute for institutional protocols, however. A clinician under time pressure may defer to the gauge without engaging critically with the uncertainty interval or the SHAP attribution. Preventing automation bias requires governance structures around tool deployment, not only interface design.

6.6.2 Biased recommendations and underrepresented groups

TCGA-OV's racial composition reflects known disparities in genomic research participation: White patients dominate the cohort and minority groups are small (Landry et al., 2018). The subgroup PEHE analysis does not reveal a systematic penalty for the race minority group, but this is weak evidence of fairness, not strong evidence of its absence. The encoder's prognostic representations may be less calibrated for minority patients in ways that a five-variable analysis cannot detect. Deployment in diverse clinical settings would require local-population auditing before use.

6.6.3 Binary treatment and clinical completeness

A clinician acting on a binary positive recommendation may receive a recommendation that is locally correct (initiate platinum) but globally incomplete (specific regimen, dose, scheduling). CaML-OP cannot flag this incompleteness.

6.6.4 Patient privacy

CaML-OP was developed on de-identified TCGA data under NCI's data-access policies. Any clinical deployment would draw the five inputs from institutional patient records under applicable data-protection regulation (GDPR in the EU). The minimal five-covariate set is chosen partly for interpretability and partly because it reduces re-identification risk relative to omics-based approaches.

6.6.5 Non-deployment status

CaML-OP is a research prototype. It has not undergone clinical validation, conformity assessment under Regulation (EU) 2024/1689, CE marking under the EU Medical Devices Regulation, or FDA clearance. No treatment decisions should be based on its outputs. The contribution of the thesis is methodological; translation to clinical use would require a separate, multi-year programme.

6.7 Threats to validity

Following Wohlin et al. (2012), threats are organized into four categories.

6.7.1 Internal validity

- **DGP assumptions.** The benchmark covers four heterogeneity structures but not the true platinum-response surface, which likely involves BRCA status and proteomic variables absent here. Rankings could shift under more biologically realistic DGPs.
- **TCGA selection bias.** TCGA patients were recruited at academic medical centres, systematically differing from community-hospital populations. The conditional unconfoundedness assumption is harder to defend in this context.
- **Hyperparameter uniformity.** All Random Forest models use a common configuration (200 trees, max depth 6, min samples 5). Separately tuning each method could change rankings, but would confound meta-learning strategy with hyperparameter optimization.
- **Encoder leakage.** The outcome-trained encoder produces leaf embeddings before causal cross-fitting begins, leaking some outcome information into the covariate space.

6.7.2 External validity

- **Generalizability.** Results on TCGA-OV and the binary five-year outcome may not transfer to other cancer types, treatment modalities, or outcome definitions.
- **Temporal coverage.** Diagnoses from 1992 to 2013 may not reflect contemporary treatment-response patterns.

6.7.3 Construct validity

- **Binary survival outcome.** Five-year binary survival ignores time-to-event structure, quality of life and toxicity.
- **PEHE as a headline metric.** PEHE rank does not predict policy quality. Reporting PEHE as the primary metric alone could mislead about clinically relevant performance; co-reporting estimated AIPW value mitigates this.
- **Adapted survival comparators.** The Cox PH and RF classifiers in this benchmark are binary-outcome predictive comparators rather than full survival baselines. They should not be read as direct evidence for or against full Cox PH or full RSF performance on time-to-event ovarian cancer data.

6.7.4 Conclusion validity

- **Test-set size.** With `n_test` approximately 137 per run, per-run standard errors are non-trivial. Aggregating 40 runs helps but does not eliminate the issue.
- **No multiple-comparison correction.** Seven causal methods compared pairwise produces multiple tests. Only two comparisons of interest were pre-specified; the rest are exploratory.

Chapter 7

Conclusion

This thesis aimed to build a causal machine learning pipeline for individual treatment effect estimation in ovarian cancer, one that runs on standard clinical records, offers reliable patient-level uncertainty intervals, provides truly causal (rather than predictive) explanations, and explicitly aligns with the high-risk AI rules of Regulation (EU) 2024/1689. The resulting system is CaML-OP.

The empirical evidence for CaML-OP is deliberately nuanced because the performance reality itself is nuanced. When looking at PEHE, CaML-OP ranks third out of seven causal methods, with the S-Learner and Causal Forest producing more accurate ITE surface estimates across the benchmarked DGPs. Yet, when evaluated on off-policy AIPW estimated policy value, CaML-OP takes first place. Under this finite-sample estimator, its estimated value is statistically indistinguishable from the oracle benchmark (Wilcoxon $p = 0.42$). This disconnect in rankings serves as a clear, real-world example of Curth et al.’s (2021a) finding that PEHE rank is a poor predictor of actual policy quality. Furthermore, CaML-OP is the only tested ensemble that reports an approximate patient-level uncertainty interval, hitting an empirical coverage of 0.847 against our 0.95 nominal target. Subgroup PEHE stayed within a tight 0.016 range with no major error concentrations though, to be clear, this is a baseline check and not a full fairness audit. Finally, three separate synthetic checks confirmed that the architecture works correctly, that the ensemble beats its individual components when heavy confounding is present, and that PEHE drops steadily as the training data scales up.

Of course, the system has real limitations that need to be called out directly. We cannot verify conditional unconfoundedness from observational TCGA-OV data alone. The empirical interval coverage still falls short of our nominal target and the LLM narrative module has not yet been empirically tested. Furthermore, the system lacks prospective validation, and its architecture is merely AI Act-*informed*, not AI Act-*complete* or legally ready for deployment.

None of these caveats diminish the contribution of this work; instead, they map out the next research agenda. The logical next steps are clear: a prospective validation study on a demographically representative cohort with deeper confounder data, an extension to multi-valued treatment encodings to handle protocol-level differences, a formal usability evaluation of the dashboard with real clinicians and a rigorous empirical pass-rate analysis of the LLM narrative module.

Ultimately, the core design choices behind CaML-OP were to use minimal covariates, keeping

uncertainty transparent, prioritizing causal explanations and anchoring the architecture in the EU AI Act aim to prove a broader point. We can advance equitable access to data-driven decision tools using routine clinical data, all while staying firmly aligned with the future of medical AI regulation.

References

- Athey, S., & Imbens, G. (2016). Recursive partitioning for heterogeneous causal effects. *Proceedings of the National Academy of Sciences*, 113(27), 7353-7360. <https://doi.org/10.1073/pnas.1510489113>
- Bray, F., Laversanne, M., Sung, H., Ferlay, J., Siegel, R. L., Soerjomataram, I., & Jemal, A. (2024). Global cancer statistics 2022: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. *CA: A Cancer Journal for Clinicians*, 74(3), 229-263. <https://doi.org/10.3322/caac.21834>
- Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 785-794). <https://doi.org/10.1145/2939672.2939785>
- Colombo, N., Sessa, C., du Bois, A., Ledermann, J., McCluggage, W. G., McNeish, I., & Vergote, I. (2019). ESMO-ESGO consensus conference recommendations on ovarian cancer: Pathology and molecular biology, early and advanced stages, borderline tumours and recurrent disease. *Annals of Oncology*, 30(5), 672-705. <https://doi.org/10.1093/annonc/mdz062>
- Cox, D. R. (1972). Regression models and life-tables. *Journal of the Royal Statistical Society, Series B (Methodological)*, 34(2), 187-220.
- Curth, A., Svensson, D., Weatherall, J., & van der Schaar, M. (2021a). Really doing great at estimating CATE? A critical look at ML benchmarking practices in treatment effect estimation. In *Proceedings of the 35th Conference on Neural Information Processing Systems Datasets and Benchmarks Track*.
- Curth, A., Lee, C., & van der Schaar, M. (2021b). SurvITE: Learning heterogeneous treatment effects from time-to-event data. In *Advances in Neural Information Processing Systems 34* (pp. 26740-26753).
- European Parliament and Council. (2024). Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act). *Official Journal of the European Union*, L 2024/1689. <https://eur-lex.europa.eu/eli/reg/2024/1689/oj>
- Gao, H., Korn, J. M., Ferretti, S., Monahan, J. E., Wang, Y., Singh, M., Zhang, C., Schnell, C., Yang, G., Zhang, Y., Balbin, O. A., Barbe, S., Cai, H., Casey, F., Chatterjee, S., Chiang, D. Y., Chuai, S., Cogan, S. M., Collins, S. D., ... Sellers, W. R. (2015). High-throughput screening using patient-derived tumor xenografts to predict clinical trial drug response. *Nature Medicine*, 21(11), 1318-1325. <https://doi.org/10.1038/nm.3954>
- Goddard, K., Roudsari, A., & Wyatt, J. C. (2012). Automation bias: A systematic review of frequency, effect mediators, and mitigators. *Journal of the American Medical Informatics*

- Association, 19(1), 121-127. <https://doi.org/10.1136/amiajnl-2011-000089>
- Hanker, L. C., Loibl, S., Burchardi, N., Pfisterer, J., Meier, W., Pujade-Lauraine, E., Ray-Coquard, I., Sehouli, J., Harter, P., & du Bois, A. (2012). The impact of second to sixth line therapy on survival of relapsed ovarian cancer after primary taxane/platinum-based therapy. *Annals of Oncology*, 23(10), 2605-2612. <https://doi.org/10.1093/annonc/mds203>
- Holland, P. W. (1986). Statistics and causal inference. *Journal of the American Statistical Association*, 81(396), 945-960. <https://doi.org/10.1080/01621459.1986.10478354>
- Imbens, G. W., & Rubin, D. B. (2015). Causal inference for statistics, social, and biomedical sciences: An introduction. Cambridge University Press.
- Ishwaran, H., Kogalur, U. B., Blackstone, E. H., & Lauer, M. S. (2008). Random survival forests. *Annals of Applied Statistics*, 2(3), 841-860. <https://doi.org/10.1214/08-AOAS169>
- Kennedy, E. H. (2023). Towards optimal doubly robust estimation of heterogeneous causal effects. *Electronic Journal of Statistics*, 17(2), 3008-3049. <https://doi.org/10.1214/23-EJS2151>
- Kilim, O., Olar, A., Biricz, A., Madaras, L., Pollner, P., Szallasi, Z., Sztupinszki, Z., & Csabai, I. (2025). Histopathology and proteomics are synergistic for high-grade serous ovarian cancer platinum response prediction. *npj Precision Oncology*, 9, 27. <https://doi.org/10.1038/s41698-025-00808-w>
- Kunzel, S. R., Sekhon, J. S., Bickel, P. J., & Yu, B. (2019). Metalearners for estimating heterogeneous treatment effects using machine learning. *Proceedings of the National Academy of Sciences*, 116(10), 4156-4165. <https://doi.org/10.1073/pnas.1804597116>
- Landry, L. G., Ali, N., Williams, D. R., Rehm, H. L., & Bonham, V. L. (2018). Lack of diversity in genomic databases is a barrier to translating precision medicine research into practice. *Health Affairs*, 37(5), 780-785. <https://doi.org/10.1377/hlthaff.2017.1595>
- Louizos, C., Shalit, U., Mooij, J., Sontag, D., Zemel, R., & Welling, M. (2017). Causal effect inference with deep latent-variable models. In *Advances in Neural Information Processing Systems* 30 (pp. 6446-6456).
- Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems* 30 (pp. 4765-4774).
- Lundberg, S. M., Erion, G., Chen, H., DeGrave, A., Prutkin, J. M., Nair, B., Katz, R., Himmelfarb, J., Bansal, N., & Lee, S.-I. (2020). From local explanations to global understanding with explainable AI for trees. *Nature Machine Intelligence*, 2(1), 56-67. <https://doi.org/10.1038/s42256-019-0138-9>
- Nie, X., & Wager, S. (2021). Quasi-oracle estimation of heterogeneous treatment effects. *Biometrika*, 108(2), 299-319. <https://doi.org/10.1093/biomet/asaa076>
- Obermeyer, Z., & Emanuel, E. J. (2016). Predicting the future - big data, machine learning, and clinical medicine. *New England Journal of Medicine*, 375(13), 1216-1219. <https://doi.org/10.1056/NEJMp1606181>
- Obermeyer, Z., Powers, B., Vogeli, C., & Mullainathan, S. (2019). Dissecting racial bias in an algorithm used to manage the health of populations. *Science*, 366(6464), 447-453. <https://doi.org/10.1126/science.aax2342>
- Pauli, C., Hopkins, B. D., Prandi, D., Shaw, R., Fedrizzi, T., Sboner, A., Sailer, V., Augello, M., Puca, L., Rosati, R., McNary, T. J., Churakova, Y., Cheung, C., Triscott, J., Pisapia, D.,

- Rao, R., Mosquera, J. M., Robinson, B., Faltas, B. M., ... Rubin, M. A. (2017). Personalized in vitro and in vivo cancer models to guide precision medicine. *Cancer Discovery*, 7(5), 462-477. <https://doi.org/10.1158/2159-8290.CD-16-1154>
- Pearl, J. (2009). *Causality: Models, reasoning, and inference* (2nd ed.). Cambridge University Press.
- Prigerson, H. G., Bao, Y., Shah, M. A., Paulk, M. E., LeBlanc, T. W., Schneider, B. J., Garrido, M. M., Reid, M. C., Berlin, D. A., Adelson, K. B., Neugut, A. I., & Maciejewski, P. K. (2015). Chemotherapy use, performance status, and quality of life at the end of life. *JAMA Oncology*, 1(6), 778-784. <https://doi.org/10.1001/jamaoncol.2015.2378>
- Reyes, M., Meier, R., Pereira, S., Silva, C. A., Dahlweid, F.-M., von Tengg-Kobligh, H., Summers, R. M., & Wiest, R. (2020). On the interpretability of artificial intelligence in radiology: Challenges and opportunities. *Radiology: Artificial Intelligence*, 2(3), e190043. <https://doi.org/10.1148/ryai.2020190043>
- Robinson, P. M. (1988). Root-N-consistent semiparametric regression. *Econometrica*, 56(4), 931-954. <https://doi.org/10.2307/1912705>
- Rubin, D. B. (1974). Estimating causal effects of treatments in randomized and nonrandomized studies. *Journal of Educational Psychology*, 66(5), 688-701. <https://doi.org/10.1037/h0037350>
- Shalit, U., Johansson, F. D., & Sontag, D. (2017). Estimating individual treatment effect: Generalization bounds and algorithms. In *Proceedings of the 34th International Conference on Machine Learning* (pp. 3076-3085).
- Shenoy, N., Bhagat, T. D., Bhagat, A., & Verma, A. (2020). Predicting ovarian cancer patients' clinical response to platinum-based chemotherapy by their tumor proteomic signatures. *Journal of Proteome Research*, 19(10), 4194-4206. <https://doi.org/10.1021/acs.jproteome.0c00525>
- Siegel, R. L., Giaquinto, A. N., & Jemal, A. (2024). Cancer statistics, 2024. *CA: A Cancer Journal for Clinicians*, 74(1), 12-49. <https://doi.org/10.3322/caac.21820>
- Tayebi Arasteh, S., Han, T., Lotfinia, M., Kuhl, C., Kather, J. N., Truhn, D., & Nebelung, S. (2024). Large language models streamline automated machine learning for clinical studies. *Nature Communications*, 15, 1603. <https://doi.org/10.1038/s41467-024-45879-8>
- TCGA Research Network. (2011). Integrated genomic analyses of ovarian carcinoma. *Nature*, 474(7353), 609-615. <https://doi.org/10.1038/nature10166>
- Thirunavukarasu, A. J., Ting, D. S. J., Elangovan, K., Gutierrez, L., Tan, T. F., & Ting, D. S. W. (2023). Large language models in medicine. *The Lancet Digital Health*, 5(7), e384-e395. [https://doi.org/10.1016/S2589-7500\(23\)00080-5](https://doi.org/10.1016/S2589-7500(23)00080-5)
- van Buuren, S., & Groothuis-Oudshoorn, K. (2011). mice: Multivariate imputation by chained equations in R. *Journal of Statistical Software*, 45(3), 1-67. <https://doi.org/10.18637/jss.v045.i03>
- Wager, S., & Athey, S. (2018). Estimation and inference of heterogeneous treatment effects using random forests. *Journal of the American Statistical Association*, 113(523), 1228-1242. <https://doi.org/10.1080/01621459.2017.1319839>
- Wohlin, C., Runeson, P., Host, M., Ohlsson, M. C., Regnell, B., & Wesslen, A. (2012). *Experimentation in software engineering*. Springer.
- Wright, A. A., Keating, N. L., Ayanian, J. Z., Chrischilles, E. A., Kahn, K. L., Ritchie, C. S.,

Weeks, J. C., Earle, C. C., & Landrum, M. B. (2016). Family perspectives on aggressive cancer care near the end of life. *JAMA*, 315(3), 284-292. <https://doi.org/10.1001/jama.2015.18604>

Yin, C., Yang, J., Wang, F., & Sun, S. (2022). Prediction model of platinum response and prognosis in ovarian cancer based on multi-omics data. *Frontiers in Oncology*, 12. (Manual verification recommended at submission time.)